6.1 Affine Varieties

In order to generalise the usual notions to non-algebraically closed field, and develop a "coordinate-free" approach, we introduce the following concept

Definition 6.1.1
Let A be a finitely generated k-algebra and $\mathfrak{m}$ a maximal ideal. Recall (3.34.21) that $k(\mathfrak{m})/k$ is an algebraic (indeed finite) field extension, and we call it the ***residue field*** *for $\mathfrak{m}$.*

Now we may introduce the Zero-Loci and study relationships to ideals

Proposition 6.1.2 (Correspondence between Zero-Loci and Ideals)
Let $A = k[X_1, ..., X_n]$ be the polynomial ring in n-variables over a field k. For a set $S \subseteq k[X_1, ..., X_n]$ and an algebraic extension K/k define the ***zero-locus***

$$V_K(S) \coloneqq \{\alpha \in K^n \mid f(\alpha) = 0 \quad \forall f \in S\}.$$

Similarly for a subset $Y \subseteq K^n$ define

$$I_k(Y) \coloneqq \{f \in A \mid f(y) = 0 \quad \forall y \in Y\}$$

The pair of maps V_K, I_k constitute a Galois Connection

$$\mathcal{I}(k[X_1, ..., X_n]) \underset{I_k}{\overset{V_K}{\rightleftharpoons}} \mathcal{P}(K^n)$$

namely they satisfy

a) V_K and I_k are order-reversing
b) $S \subseteq I_k(V_K(S))$
c) $Y \subseteq V_K(I_k(Y))$

Furthermore (omitting the subscripts)

d) $VIV = V$ and $IVI = I$
e) $I(Y)$ is a radical ideal and $\sqrt{\langle S\rangle} \subseteq I(V(S))$
f) $V(S) = V(\langle S\rangle) = V\left(\sqrt{\langle S\rangle}\right)$ and $V(\mathfrak{a}) = V\left(\sqrt{\mathfrak{a}}\right)$
g) $\bigcap_i V(S_i) = V\left(\bigcup_i S_i\right)$ and $\bigcap_i V(\mathfrak{a}_i) = V\left(\sum_i \mathfrak{a}_i\right)$
h) $\bigcap_i I(W_i) = I\left(\bigcup_i W_i\right)$
i) $V(\mathfrak{a}) \cup V(\mathfrak{b}) = V(\mathfrak{a} \cap \mathfrak{b}) = V(\mathfrak{a}\mathfrak{b})$
j) $V((0)) = K^n$ and $V(A) = \emptyset$

The sets of the form $V_K(\mathfrak{a})$ constitute the closed sets of a topology on K^n, denoted by $\mathrm{Zar}_k(K^n)$. In this case we have the following form for the topological closure

$$V_K(I_k(Y)) = \overline{Y}$$

Furthermore

$$I_k(Y) = I_k\left(\overline{Y}\right)$$

Proof. We make use of general results on Galois connections (Foundations, Set Theory), though many results may be shown more directly. The fact it's a Galois connection follows from (2.1.54).

1-3. These follow (2.1.50)

4. This follows from (2.1.52).

5. It's clear that $I(Y)$ is an ideal. It is radical because K is reduced (...). The second statement is straightforward.

6. This follows from (2.1.53) by considering the closure operators $\sqrt{\langle -\rangle}$ and $\langle -\rangle$.

7. The first equality follows from (2.1.55). The second equality follows from (3.3.29).

8. This follows from (2.1.55).

9. Observe that $\mathfrak{m}_x$ is prime (because K is an integral domain) and $x \in V(\mathfrak{a}) \iff \mathfrak{a} \subseteq \mathfrak{m}_x$. the result follows from (3.3.38) because $\mathfrak{a} \subseteq \mathfrak{m}_x \vee \mathfrak{b} \subseteq \mathfrak{m}_x \iff \mathfrak{a}\mathfrak{b} \subseteq \mathfrak{m}_x \iff \mathfrak{a} \cap \mathfrak{b} \subseteq \mathfrak{m}_x$

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The family of sets $\mathrm{Zar}_k(K^n) := \mathrm{Im}(V_K)$ constitute the closed sets of a topology precisely because they are closed under arbitrary intersections and finite unions. Furthermore by (2.1.52) $V_K \circ I_k$ is a closure operator with image precisely the closed sets. Therefore by (2.1.41)

$$(V_K \circ I_k)(Y) = \bigcap_{Y \subseteq Z \in \mathrm{Zar}_k(K^n)} Z$$

which is the definition of the topological closure. The last statement follows by applying I_k and using **4.** ∎

For a fixed ideal $\mathfrak{a} \lhd k[X_1, ..., X_n]$ we may vary the field K to obtain families of solutions in different fields.

Definition 6.1.3 (Algebraic Set and L-valued points)
*Let $\mathfrak{a} \lhd k[X_1, ..., X_n]$ be a **radical** ideal. For every k-algebra R define the R-valued points by*

$$V(\mathfrak{a})(R) := \{r \in R^n \mid f(r_1, ..., r_n) \quad \forall f \in \mathfrak{a}\}$$

Definition 6.1.4 (Affine Variety)
*Let $\mathfrak{a} \lhd k[X_1, ..., X_n]$ be a **radical** ideal. The family $X(-) := V(\mathfrak{a})(-)$ is called an **affine variety** to which we associate the **coordinate ring** $k[X] := k[X_1, ..., X_n]/\mathfrak{a}$, which is a reduced f.g. k-algebra. Note the elements of $k[X]$ may be regarded as R-valued functions on the R-valued points $X(R)$.*

Note we may regard this as a functor, suppose we have compatible maps $i_{jk} : K_j/k \to K_k/k$ then these induce compatible maps

$$X(i_{jk}) : X(K_j) \to X(X_k)$$

Definition 6.1.5 (Residue Field)
*Let $X = V(\mathfrak{a})$ be an affine variety and L/k a field extension. For a point $x \in X(L)$ define the **residue field** to be*

$$k(x) := k(x_1, ..., x_n) \subset L/k$$

and the degree of x to be

$$\deg(x) := [k(x) : k]$$

Furthermore we define

$$\mathfrak{M}_{X,x} := I_k(\{(x)\})$$

for $x \in X(L)$. When x is algebraic then $\deg(x)$ is finite and $\mathfrak{M}_{X,x} = \mathfrak{m}_x/\mathfrak{a}$ is maximal (3.34.21).

Remark 6.1.6
$\mathbb{A}^n_k := V((0))$ is an affine variety and $\mathbb{A}^n_k(L) = L^n$.

Proposition 6.1.7 (Rational Points are Algebra Homomorphisms)
Let $X = V(\mathfrak{a})$ be an affine variety and R a k-algebra. Then there is a bijection

$$\{x \in X(R)\} \longleftrightarrow \mathrm{AlgHom}_k(k[X], R)$$

Proof. Given $x \in X(R)$ then the evaluation map $\mathrm{ev}_x : k[X_1, ..., X_n] \to R$ contains $\mathfrak{a}$ by definition and therefore (3.10.3) yields a unique map $\varphi_x : k[X] \to R$ such that $\varphi_x\left(\overline{X}_i\right) = x_i$.

Similarly given an algebra homomorphism $\varphi : k[X] \to R$ we may define $x = \left(\varphi\left(\overline{X}_1\right), ..., \varphi\left(\overline{X}_n\right)\right)$. Then by uniqueness these are mutual inverses. ∎

For completeness we also consider affine subvarieties in the same way as before

Proposition 6.1.8 (Affine subvarieties)
*Let $X = V(\mathfrak{a}) \subset \mathbb{A}^n_k$ be an affine variety and $A := k[X]$ the coordinate ring. For any ideal $\mathfrak{b} \lhd A$ and field extension K/k define the **zero-locus** by*

$$V_K(\mathfrak{b}) := \{(x) \in X(K) \mid f(x) = 0 \quad \forall f \in \mathfrak{b}\}$$

Similarly for a subset $Y \subset X(K)$ define

$$I_k(Y) := \{f \in k[X] \mid f(x) = 0 \quad \forall x \in Y\}$$

Suppose $\pi : k[X_1, ..., X_n] \to k[X]$ is the quotient map. Then we have the following properties

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$$V_K(\mathfrak{b}) = V_K(\pi^{-1}(\mathfrak{b})) \cap X(K)$$
$$I_k(Y \cap X(K)) = \pi(I_k(Y)) \quad Y \subset \mathbb{A}_k^n(K)$$
$$I_k(V_K(\mathfrak{b})) = \pi(I_k(V_K(\pi^{-1}(\mathfrak{b}))))$$

The pair (V_K, I_k) satisfies the same properties as (...). The image of V_K forms the closed sets of a topology on $X(K)$ which coincides with the subspace topology from $\mathbb{A}_k^n(K)$.

We may now generalize the Weak Nullstellensatz to arbitrary affine varieties and non-algebraically closed fields K/k.

Proposition 6.1.9 (Galois Group Action)
Let $X = V(\mathfrak{a})$ be an affine variety and K/k a normal *algebraic field extension (e.g. $\overline{k}$). There is a well-defined group action*

$$\mathrm{Aut}(K/k) \times X(K) \to X(K)$$
$$(\sigma, (x_1, ..., x_n)) \to (\sigma(x_1), ..., \sigma(x_n))$$

For $x, y \in X(K)$ the following are equivalent

a) x and y are topologically indistinguishable
b) $x = \sigma(y)$ for some $\sigma \in \mathrm{Aut}(K/k)$
c) $x \in \overline{\{y\}}$
d) $y \in \overline{\{x\}}$
e) $\overline{\{x\}} = \overline{\{y\}}$
f) $\mathfrak{M}_{X,x} = \mathfrak{M}_{X,y}$

For every point x we have

$$V_K(\mathfrak{M}_{X,x}) = \overline{\{x\}} = \{\sigma(x) \mid \sigma \in \mathrm{Aut}(K/k)\}$$

and x is quasi-closed (4.1.90) and therefore $X(K)$ is symmetric. Denote the set of Galois Orbits by $X_0(K)$ then the quotient map

$$\pi : X(K) \to X_0(K)$$

is precisely the Kolmogorov Quotient (4.1.36) and in particular a quasi-homeomorphism.

Proof. Given $F \in \mathfrak{a}$ we have $\sigma(F(x)) = F(\sigma(x))$. This shows that $x \in X(K) \implies \sigma(x) \in X(K)$. Further it's clear that $\sigma(\tau(x)) = (\sigma \circ \tau)(x)$ so the group action is well-defined.

By (4.1.34) a) $\iff$ e) $\implies$ c), d). By (6.1.2) $V(\mathfrak{M}_{X,x}) = \overline{\{x\}}$ and $I(\overline{\{x\}}) = \mathfrak{M}_{X,x}$. So e) $\iff$ f). Suppose $\mathfrak{M}_{X,x} = \mathfrak{M}_{X,y}$ then this means there exists an isomorphism $k(x) \cong k(y) \subset K$ which lifts to an element of $\mathrm{Aut}(K/k)$ by (3.18.69), so f) $\implies$ b). Conversely b) $\implies$ f) is straightforward. If $x \in \overline{\{y\}} = V(\mathfrak{M}_{X,y})$ then $\mathfrak{M}_{X,y} \subset \mathfrak{M}_{X,x}$ whence they are equal by maximality, so c), d) $\implies$ f). ∎

Corollary 6.1.10
Let k be an algebraically closed field and X an affine variety. Then $X(k)$ is a Kolmogorov topological space.

Proposition 6.1.11 (Generalized Weak Nullstellensatz)
Let $X = V(\mathfrak{a})$ be an affine variety. There is a bijection between maximal ideals and Galois orbits of points

$$\mathrm{Specm}(k[X]) \longleftrightarrow X_0(\overline{k}) = X(\overline{k})/\mathrm{Aut}(\overline{k}/k)$$
$$\mathfrak{M} \to [V_K(\mathfrak{M})]$$
$$\mathfrak{M}_{X,x} \leftarrow [x]$$

Proof. We may deduce this from (3.34.24) by observing that $x \in X(\overline{k}) \iff \mathfrak{a} \subset \mathfrak{m}_x$, $\mathfrak{M}_{X,x} = \mathfrak{m}_x/\mathfrak{a}$ and using the correspondence of ideals of quotient ring (3.3.56)

$$\{\mathfrak{M} \in \mathrm{Specm}(k[X_1, ..., X_n]) \mid \mathfrak{a} \subset \mathfrak{M}\} \longleftrightarrow \mathrm{Specm}(k[X]).$$

∎

Lemma 6.1.12 (Rabinowitsch Trick)
Let $\mathfrak{a} \lhd A$ and $f \in A$. Consider the ring $B = A[Y]$. If $\mathfrak{a}B + (1 - Yf) = B$ then $f \in \sqrt{\mathfrak{a}}$.

Proof. The hypothesis implies

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$$1 = (1 - Yf)g(Y) + ah(Y)$$

for $a \in \mathfrak{a}$ and $h(Y) \in A[Y]$. Consider the quotient map $\bar{\cdot} : A \to A/\mathfrak{a}$ and the corresponding map $A[Y] \to (A/\mathfrak{a})[Y]$. Applying this to the above shows $1 - Y\overline{f}$ is invertible in $(A/\mathfrak{a})[Y]$. So by (3.7.5) $\overline{f}$ is nilpotent in $(A/\mathfrak{a})$ whence $f \in \sqrt{\mathfrak{a}}$. ∎

Proposition 6.1.13 (Strong Nullstellensatz)
Let X be an affine variety and $\mathfrak{a} \lhd k[X]$. Then

$$I_k V_{\overline{k}}(\mathfrak{a}) = \sqrt{\mathfrak{a}}^J = \sqrt{\mathfrak{a}}$$

In particular the $\overline{k}$-radical ideals are precisely the radical ideals. and there is a dual isomorphism

$$\mathrm{Rad}(k[X]) \longleftrightarrow \mathrm{Zar}_k\left(X\left(\overline{k}\right)\right)$$

Proof. Observe that $x \in V_{\overline{k}}(\mathfrak{a})$ if and only if $\mathfrak{a} \subseteq \mathfrak{M}_{X,x}$. Therefore

$$I_k(V_{\overline{k}}(\mathfrak{a})) = \bigcap_{x \in V_{\overline{k}}(\mathfrak{a})} I_k(\{x\}) = \bigcap_{\mathfrak{a} \subseteq \mathfrak{M}_{X,x}} \mathfrak{M}_{X,x}$$

Therefore by the correspondence in (6.1.11) this is equal to precisely $\sqrt{\mathfrak{a}}^J$.

For the second equality we consider first the case $X = \mathbb{A}^n_k$ and $k[X] = k[X_1, ..., X_n]$. Let $\mathfrak{a} \lhd k[X]$ and choose $f \in I_k V_{\overline{k}}(\mathfrak{a})$. Consider the ring $B \coloneqq k[X_1, ..., X_n, Y]$ and the ideal $\tilde{\mathfrak{a}} = \mathfrak{a}B + (1 - Yf)$. Clearly this has no zeros in $\overline{k}^{n+1}$, so by the Weak Nullstellensatz (3.34.24) it is not proper. By the (6.1.12) $f \in \sqrt{\mathfrak{a}}$ as required. The reverse inclusion is clear.

Now suppose that $X = V(\mathfrak{a})$, $k[X] = k[X_1, ..., X_n]/\mathfrak{a}$ and $\mathfrak{b} \lhd k[X]$ is proper. Using (6.1.8) and (3.3.50) together with the result just proven, shows

$$I_k(V_{\overline{k}}(\mathfrak{b})) = \pi(I_{\overline{k}} V_{\overline{k}}(\pi^{-1}\mathfrak{b})) = \pi\left(\sqrt{\pi^{-1}(\mathfrak{b})}\right) = \sqrt{\mathfrak{b}}$$

where $\pi : k[X_1, ..., X_n] \to k[X]$ is canonical surjective morphism. ∎

Corollary 6.1.14
Let $k[X]$ be any finitely generated reduced k-algebra, then $k[X]$ is a Jacobson ring, i.e.

$$\sqrt{\mathfrak{a}} = \sqrt{\mathfrak{a}}^J$$

In particular the intersection of all maximal ideals is zero

$$\bigcap_{\mathfrak{m}} \mathfrak{m} = 0$$

6.1.1 Topological Properties

Proposition 6.1.15 (Principal Open Sets)
Let $X = V(\mathfrak{a})$ be an affine variety over k. For every $f \in k[X]$ the sets

$$D(f) \coloneqq X\left(\overline{k}\right) \setminus V_{\overline{k}}((f)) = \left\{(x) \in X\left(\overline{k}\right) \;\middle|\; f(x) \neq 0\right\}$$

form a base for the Zariski topology $\mathrm{Zar}_k\left(X\left(\overline{k}\right)\right)$.

Proof. Let $U = X\left(\overline{k}\right) \setminus V(\mathfrak{b})\left(\overline{k}\right)$ be a non-empty open set. For every $x \in U$ there by definition exists $f \in \mathfrak{b}$ such that $f(x) \neq 0$. Therefore $D(f)$ is a neighbourhood of x in the Zariski topology. ∎

The topological notion of irreducibility is important, and may be reduced to a purely algebraic statement on the coordinate ring.

Proposition 6.1.16 (Criterion for Irreducibility)
Let X be an affine variety (e.g. $\mathbb{A}^n_k$) and $Y = V(\mathfrak{b})$ an affine subvariety corresponding to a radical ideal $\mathfrak{b} \lhd k[X]$. Then the following are equivalent

a) $Y\left(\overline{k}\right)$ is an irreducible subset of $X\left(\overline{k}\right)$
b) $Y_0\left(\overline{k}\right)$ is an irreducible subset of $X_0\left(\overline{k}\right)$
c) $\mathfrak{b}$ is prime
d) $k[Y]$ is an integral domain.

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Proof. Suppose X is not irreducible. Then we have $X \subseteq V_{\overline{k}}(\mathfrak{b}) \cup V_{\overline{k}}(\mathfrak{c})$ a non-trivial decomposition into closed subsets (and associated ideals). Then by the dual isomorphism we have also $\mathfrak{a} \subsetneq \mathfrak{b}$ and we may choose $f \in \mathfrak{b} \setminus \mathfrak{a}$ and similarly $g \in \mathfrak{c} \setminus \mathfrak{a}$. However fg vanishes on $X\left(\overline{k}\right)$ ands so we have $fg \in \mathfrak{a}$. Therefore $\mathfrak{a}$ is not prime.

Conversely suppose X is irreducible and $\mathfrak{bc} \subseteq \mathfrak{a}$. Then $X \subseteq V_{\overline{k}}(\mathfrak{b}) \cup V_{\overline{k}}(\mathfrak{c})$. By irreducibility we have $X \subseteq V_{\overline{k}}(\mathfrak{b})$, whence applying $I_{\overline{k}}(-)$ we see $\mathfrak{b} \subseteq I_{\overline{k}}V_{\overline{k}}(\mathfrak{b}) \subseteq \mathfrak{a}$ (since $\mathfrak{a}$ is radical). Therefore $\mathfrak{a}$ is prime. ∎

When $W \subset X$ is an irreducible subset then write $\mathfrak{M}_{X,W} \coloneqq I(W)$, by the previous result this equals $I\left(\overline{W}\right)$ and $\mathfrak{M}_{X,W}$ is a prime ideal.

Proposition 6.1.17
Let X be an affine variety, W, Z be subsets of $X\left(\overline{k}\right)$. Then

$$W \subset \overline{Z} \iff \mathfrak{M}_{X,Z} \subset \mathfrak{M}_{X,W}$$

and

$$\overline{W} = \overline{Z} \iff \mathfrak{M}_{X,Z} = \mathfrak{M}_{X,W}$$

Proof. Observe that $V\left(\mathfrak{M}_{X,W}\right) = \overline{W}$ by (6.1.2). Therefore the result follows immediately from the same result. ∎

Proposition 6.1.18 (Points are Closed)
Let $X = V(\mathfrak{a})$ be an affine variety and $x \in X\left(\overline{k}\right)$. Then

$$\overline{\{x\}} = V\left(\mathfrak{M}_{X,x}\right) = \left\{\sigma(x) \;\middle|\; \sigma \in \mathrm{Aut}\left(\overline{k}/k\right)\right\}$$

is irreducible and finite of order $[k(x) : k]_s$. Further X is a symmetric space.

Proof. The equalities are proven in (6.1.9). Further $\overline{\{x\}}$ is irreducible by (4.1.67). We may replace $\mathrm{Aut}\left(\overline{k}/k\right)$ with $\mathrm{Mor}_k\left(k(x), \overline{k}\right)$ to determine the order.

Clearly X is symmetric because it satisfies criteria (4.1.41) b). ∎

Proposition 6.1.19 (Closed sets $\leftrightarrow$ radical ideals)
Let $X = V(\mathfrak{a})$ be an affine variety. Recall (6.1.13) there is a dual lattice isomorphism between radical ideals and "Zariski"-closed subsets of $X\left(\overline{k}\right)$ (and indeed $X_0\left(\overline{k}\right)$).

$$\mathrm{Rad}(k[X]) \underset{I_k(-)}{\overset{V_{\overline{k}}(-)}{\rightleftarrows}} \mathrm{Zar}_k\left(X\left(\overline{k}\right)\right)$$

Under this isomorphism we have

- *maximal ideals correspond to $\mathrm{Aut}\left(\overline{k}/k\right)$-orbits of $X\left(\overline{k}\right)$ (or elements of $X_0\left(\overline{k}\right)$)*
- *prime ideals of $k[X]$ correspond to irreducible subsets of $X\left(\overline{k}\right)$ (and therefore of $X_0\left(\overline{k}\right)$)*
- *minimal prime ideals of $k[X]$ correspond to irreducible components of $X\left(\overline{k}\right)$ (and therefore of $X_0\left(\overline{k}\right)$)*

Recall that prime ideals are precisely the meet-prime radical ideals and irreducible subsets are precisely the join-prime closed subsets (see (4.1.61)). Therefore we have a dual isomorphism between the Krull Lattice of radical ideals of $k[X]$ and the Krull Latice of closed subsets of $X\left(\overline{k}\right)$.

Proof. The content of the Strong Nullstellensatz is precisely that $I_k(-) \circ V_{\overline{k}}(-) = \mathbf{1}$. The other direction was already proven so we have a dual order isomorphism. The statement about maximal ideals was already shown in (6.1.11) and prime ideals in (6.1.16). Then as irreducible components are precisely maximal irreducible subsets the final statement follows from the dual order isomorphism. ∎

Corollary 6.1.20
Let $X = V(\mathfrak{a})$ be an affine variety. There is a bijective correspondence between affine subvarieties of X and closed subsets of $X\left(\overline{k}\right)$.

Definition 6.1.21 (Irreducible Affine Variety)
*We say an affine variety $X = V(\mathfrak{a})$ is **integral** or **irreducible** if the topological space $X\left(\overline{k}\right)$ is irreducible.*

This is the case precisely when $\mathfrak{a}$ is prime, or when $k[X]$ is an integral domain by (6.1.19), or equivalently irreducible (3.3.67), since it is assumed to be reduced.

Note trivially that $\mathbb{A}^n_k$ is irreducible.

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Proposition 6.1.22 (Decomposition into Irreducible Components)
Let $X = V(\mathfrak{a})$ be an affine variety then the topological space $X\left(\overline{k}\right)$ is Noetherian. Furthermore it has finitely many irreducible components X_i and the decomposition

$$X\left(\overline{k}\right) = X_1 \cup ... \cup X_n$$

is the unique incomparable decomposition into irreducible closed subsets.

Proof. By Hilbert's Basis Theorem (3.14.8) $k[X]$ is a Noetherian ring, so by (6.1.19) $X\left(\overline{k}\right)$ is Noetherian. The result then follows from (4.1.76) ∎

Proposition 6.1.23 (Subspace Topology)
Let $X = V(\mathfrak{a})$ an affine variety and $Y = V(\mathfrak{b})$ an affine subvariety. Then there is a commutative diagram

$$\begin{array}{ccc} \{\mathfrak{c} \in \mathrm{Rad}(k[X]) \mid \mathfrak{b} \subseteq \mathfrak{c}\} & \underset{I}{\overset{V}{\rightleftarrows}} & \left\{Z \in \mathrm{Zar}_k\left(X\left(\overline{k}\right)\right) \;\middle|\; Z \subseteq Y\left(\overline{k}\right)\right\} \\ \pi^{-1} \uparrow \;\downarrow \pi & & \| \\ \mathrm{Rad}(k[X]/\mathfrak{b}) & \underset{I}{\overset{V}{\rightleftarrows}} & \mathrm{Zar}_k\left(Y\left(\overline{k}\right)\right) \end{array}$$

under which prime ideals correspond to irreducible subsets and the horizontal arrows induce dual isomorphisms.

In particular the subspace topology for $Y\left(\overline{k}\right)$ coincides with the Zariski topology.

Proof. The left hand arrows are mutual inverses by (3.3.56). The horizontal maps are dual isomorphisms by (6.1.19). The equality then follows from (6.1.8). ∎

Proposition 6.1.24
Let X be an affinve variety. The correspondence between points and ideals (6.1.11) restricts as follows

$$\mathrm{Specm}\left(k[X]_f\right) \longleftrightarrow \{\mathfrak{M} \in \mathrm{Specm}(k[X]) \mid f \notin \mathfrak{M}\} \longleftrightarrow D(f)_0 := D(f)/\operatorname{Aut}\left(\overline{k}/k\right)$$

More precisely a point $x \in D(f)$ corresponds to the maximal ideal $\left(\mathfrak{M}_{X,x}\right)_f$ of $k[X]_f$.

Proof. For the first correspondence consider the localisation map $i : k[X] \to k[X]_f$. By (3.5.27) the extension of a maximal ideal not containing f is maximal. Conversely i satisfies the hypotheses of (3.34.27) so the inverse image of a maximal ideal is maximal. This shows the inverse image of a maximal ideal is maximal, not containing f. Therefore the correspondence of prime ideals (3.5.36) restricts to maximal ideals.

The second correspondence is immediate from (6.1.11) because $f \notin \mathfrak{M}_{X,x} \iff f(x) \neq 0$. ∎

6.1.2 Structure Sheaf

Definition 6.1.25 (Sheaf of Regular Functions)
*Let X be an affine variety, $U \subset X\left(\overline{k}\right)$ an open subset and $f : U \to \overline{k}$. We say that f is **regular** if for every $x \in U$ there exists $g, h \in k[X]$ and an open neighbourhood V of x such that*

$$f(y) = \frac{g(y)}{h(y)} \quad \forall y \in V$$

*We denote the **structure sheaf** by*

$$\mathcal{O}_X(U) := \left\{f : U \to \overline{k} \;\middle|\; f \text{ regular}\right\}$$

which is a sheaf of k-algebras.

If $U \subset X$ is an open subset then we denote the structure sheaf by the restriction

$$\mathcal{O}_U := \mathcal{O}_X \mid_U$$

Proposition 6.1.26 (Sections are localisation of coordinate ring)
Let $X = V(\mathfrak{a})$ be an affine variety and $f \in k[X]$. There is a canonical isomorphism

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$$\rho_f : k[X]_f \xrightarrow{\sim} \mathcal{O}_X(D(f))$$
$$\frac{g}{f^n} \to \frac{g(-)}{f(-)^n}$$

Furthermore $D(g) \subseteq D(f) \iff S_f \subseteq \overline{S_g}$ *and we have the following commutative diagram*

$$\begin{array}{ccc} k[X]_f & \xrightarrow{\sim} & \mathcal{O}_X(D(f)) \\ \downarrow{\scriptstyle i_{S_f S_g}} & & \downarrow{\scriptstyle (-)|_{D(g)}} \\ k[X]_g & \xrightarrow{\sim} & \mathcal{O}_X(D(g)) \end{array}$$

In particular there is a canonical isomorphim

$$k[X] \xrightarrow{\sim} \mathcal{O}_X(X)$$

Proof. The map is trivially well-defined and injective. Consider a regular map $\sigma \in \mathcal{O}_X(D(f))$. Consider the ideal

$$\mathfrak{a} \coloneqq \left\{ g \in k[X] \mid g\sigma \in \mathrm{Im}\left(i_f\right) \right\}$$

It is enough to show $f \in \mathfrak{a}$. Suppose $f \notin \mathfrak{a}$ then it's contained in a maximal ideal which is of the form $\mathfrak{M}_{X,x}$ for some $x \in X\left(\overline{k}\right)$ by (6.1.11). By definition $x \in D(f)$ and there is an open neighbourhood $W \subseteq D(f)$ and elements $h_1, h_2 \in k[X]$ such that

$$\sigma(y) = h_1 \frac{y}{h_2}(y) \quad \forall y \in W$$

Choose $h_3 \in k[X]$ such that $x \in D(h_3) \subseteq W$ then clearly

$$\sigma(y) = (h_1 h_3)\frac{y}{h_2 h_3}(y) \quad \forall y \in D(h_3)$$

and in particular $(h_2 h_3 \sigma)(y) = (h_1 h_3)(y)$ for all $y \in D(f)$. Therefore $h_2 h_3 \in \mathfrak{a} \subseteq \mathfrak{m}_x$ which implies $h_2(x) = 0$ or $h_3(x) = 0$ a contradiction.

Therefore $f \in \mathfrak{a}$ and clearly $\sigma \in \mathrm{Im}\left(i_f\right)$ as required. ∎

Corollary 6.1.27
Suppose $f \in k[X]$ *is a regular function then* $f \neq 0 \implies D(f) \neq \emptyset$.

In what follows we may systematically identify $\mathcal{O}_X(X)$ and $k[X]$, and we formalise this in the following result.

Proposition 6.1.28 (Properties of the Structure Sheaf)
Let $X = V(\mathfrak{a})$ *be an affine variety and let* $\widehat{\cdot} : k[X] \xrightarrow{\sim} \mathcal{O}_X(X)$ *be the canonical isomorphism defined in (6.1.26). Then the following properties hold*

a) For all $f \in \mathcal{O}_X(X)$ *we have* $D(f) = \{x \in U \mid f(x) \neq 0\}$ *is open and* $f|_{D(f)}$ *is invertible.*
b) For an irreducible subset $W \subset X$ *define the prime ideal*

$$\mathfrak{M}_{X,W} \coloneqq \{f \in \mathcal{O}_X(X) \mid f(x) = 0 \quad \forall x \in W\}$$

c) For $x \in X$ *we have* $\mathfrak{M}_{X,x}$ *is maximal and every maximal ideal is of this form*
d) For $W, Z \subset X$ *be irreducible subsets then*

$$\mathfrak{M}_{X,Z} \subset \mathfrak{M}_{X,W} \iff W \subset \overline{Z}$$

and

$$\mathfrak{M}_{X,Z} = \mathfrak{M}_{X,W} \iff \overline{W} = \overline{Z}$$

In particular $\mathfrak{M}_{X,W} = \mathfrak{M}_{X,\overline{W}}$
e) For all $f \in \mathcal{O}_X(X)$ *we have a canonical isomorphism*

$$\begin{array}{ccc} \mathcal{O}_X(X) & & \\ \downarrow & \searrow^{\rho_{X D(f)}} & \\ \mathcal{O}_X(X)_f & \xrightarrow{\sim} & \mathcal{O}_X(D(f)) \end{array}$$

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which is the unique homomorphism making this diagram commute

For X an affine variety we denote by $\overline{X}_1, ..., \overline{X}_n$ the image of the coordinate functions under $\hat{\cdot}$.

6.1.3 Regular Morphisms of Affine Varieties

Definition 6.1.29 (Regular Morphism of Affine Varieties)
*Let $X \subset \mathbb{A}_k^n$, $Y \subset \mathbb{A}_k^m$ be affine varieties and $\phi : X\left(\overline{k}\right) \to Y\left(\overline{k}\right)$ a function. Then we say that ϕ is a **regular morphism** (at x) if $\pi_j \circ \phi$ is regular (at x) in the sense of (6.1.25) for $j = 1, ..., m$.*

A regular morphism is an isomorphism if it has a two-sided inverse which is also regular.

Note a regular function $X \to \overline{k}$ is completely equivalent to a regular morphism $X \to \mathbb{A}_k^1$.

Proposition 6.1.30 (Regular Morphisms of Affine Varieties)
Let $X = V(\mathfrak{a})$ and $Y = V(\mathfrak{b})$ be affine varieties. Then there is a bijection

$$\begin{aligned} \mathrm{AlgHom}_k(\mathcal{O}_Y(Y), \mathcal{O}_X(X)) &\longleftrightarrow \{\phi : X \to Y \mid \phi \text{ regular}\} \\ \phi^\sharp \to & \left(\phi^\sharp\left(\overline{Y}_1\right)(-), ..., \phi^\sharp\left(\overline{Y}_m\right)(-)\right) \\ \overline{G} \to G(\pi_1 \circ \phi, ..., \pi_m \circ \phi_m) \leftarrow & \phi \end{aligned}$$

Proof. Let $\phi^\sharp$ be a k-algebra homomorphism, then $\phi^\sharp\left(\overline{Y}_i\right) \in \mathcal{O}_Y(Y)$ is a regular function. Therefore by definition the first map is well-defined. Conversely $\pi_j \circ \phi$ is regular so corresponds to an element of $\phi_j \in \mathcal{O}_X(X)$ and the second map is well-defined. As ϕ is uniquely determined by $\phi^\sharp\left(Y_j\right)$ for $j = 1, ..., m$ we see that the maps are mutually inverse. ■

Lemma 6.1.31
Let $\phi : X \to Y$ be a regular morphism of affine varieties and $W \subset X$ an irreducible subset. Then

$$\left(\phi^\sharp\right)^{-1}\left(\mathfrak{M}_{X,W}\right) = \mathfrak{M}_{Y,\phi(W)} = \mathfrak{M}_{Y,\overline{\phi(W)}}$$

Proof. This follows immediately from the definitions and (6.1.28). ■

Proposition 6.1.32
Let $\phi : X \to Y$ be a regular morphism of affine varieties and $W \subset X$, $Z \subset Y$ irreducible. Then

$$Z \subset \overline{\phi(W)} \iff \left(\phi^\sharp\right)^{-1}\left(\mathfrak{M}_{X,W}\right) \subset \mathfrak{M}_{Y,Z}$$

and

$$\overline{Z} = \overline{\phi(W)} \iff \left(\phi^\sharp\right)^{-1}\left(\mathfrak{M}_{X,W}\right) = \mathfrak{M}_{Y,Z}$$

Proof. This follows from (6.1.28) and (6.1.31). ■

Proposition 6.1.33
Let $\phi : X \to Y$ be a regular morphism of affine varieties and $x \in X$, $y \in Y$ points. Then the following are equivalent

a) $y \in \overline{\{\phi(x)\}}$
b) $\phi(x) \in \overline{\{y\}}$
c) $\overline{\{\phi(x)\}} = \overline{\{y\}}$
d) $\left(\phi^\sharp\right)^{-1}\left(\mathfrak{M}_{X,x}\right) = \mathfrak{M}_{Y,y}$

When k is algebraically closed then this is equivalent to $\phi(x) = y$.

Proof. We have a)-c) are equivalent because Y is symmetric (6.1.9), (4.1.42), and then c) $\iff$ d) follows from (6.1.32). ■

Corollary 6.1.34
Let $\phi : X \to Y$ be a regular morphism of affine varieties. Then ϕ is continuous.

Proof. Observe that $\phi^{-1}(D(f)) = D(f \circ \phi) = D\left(\phi^\sharp(f)\right)$ so we are done by (6.1.15) and (4.1.27). ■

Corollary 6.1.35
Let $\phi : X \to Y$ and $\psi : Y \to Z$ be regular morphisms of affine varieties. Then $\psi \circ \phi$ is regular and

$$(\psi \circ \phi)^\sharp = \phi^\sharp \circ \psi^\sharp$$

Proposition 6.1.36 (Affine Varieties form a Category)
The collection of affine varieties together with regular morphisms form a category $\mathbf{AffVar}_k$. Furthermore there is an equivalence of categories

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$$\mathbf{AffVar}_k \xrightarrow{\sim} \mathbf{ReducedFgAlg}_k^{op}$$

Proof. By using the correspondence with k-algebra homomorphisms (6.1.30) we may show that the law of composition for regular morphisms also satisfy the axioms of a category. We have therefore shown that the assignment $X \to k[X]$ is full and faithful functor. To show it is essentially surjective consider a reduced finitely-generated k-algebra A. Then by definition $A \cong k[X_1, ..., X_n]/\mathfrak{a}$ for some ideal $\mathfrak{a}$. As A is reduced we see $\mathfrak{a}$ is radical and therefore A is the coordinate ring of the affine variety $X = V(\mathfrak{a})$ and by definition $k[X] \cong A$. ∎

Lemma 6.1.37
Let $\phi : X \to Y$ be a regular morphism of affine varieties. Suppose that $V \subset Y$ is open and $f : V \to \overline{k}$ is regular. Then $f \circ \phi : \phi^{-1}(V) \to \overline{k}$ is regular.

In other words there is a well-defined functor

$$\mathbf{AffVar}_k \to \mathbf{LocalSpaceFunctions}_k$$

which we show is full and faithful in (6.1.39).

Proposition 6.1.38 (Principal Open Sets are Affine)
Let $X = V(\mathfrak{a})$ be an affine variety and $f \in \mathcal{O}_X(X)$. Define $Y \coloneqq V(\mathfrak{a}^e + (1 - X_{n+1}F)) \subset \mathbb{A}_k^{n+1}$ where $f = \overline{F}$ and $F \in k[X_1, ..., X_n]$. Then there is an isomorphism of spaces with functions

$$\theta : \left(D(f), \mathcal{O}_X|_{D(f)}\right) \to \left(Y\left(\overline{k}\right), \mathcal{O}_Y\right)$$
$$(x_1, ..., x_n) \to \left(x_1, ..., x_n, \frac{1}{F(x_1, ..., x_n)}\right)$$

Proof. Observe $V(\mathfrak{a}^e + (1 - X_{n+1}F)) = V(\mathfrak{a}^e) \cap V(1 - X_{n+1}F)$. Therefore θ is well-defined. Conversely given $(x_1, ..., x_{n+1}) \in V(\mathfrak{a}^e + (1 - X_{n+1}F)) = V(\mathfrak{a}^e) \cap V(1 - X_{n+1}F)$, by definition $(x_1, ..., x_n) \in V(\mathfrak{a})$ and $F(x_1, ..., x_n)x_{n+1} = 1 \implies F(x_1, ..., x_n) \neq 0$. Therefore the inverse map is well-defined. Evidently this restricts to a bijection

$$D\left(\overline{G}\right) \cap D(F) \longleftrightarrow D(G) \cap V(\mathfrak{a}^e + (1 - X_{n+1}F))$$

By (6.1.24) and (4.1.27) θ is a homeomorphism.

Suppose that $V \subset Y$ is open and $g : V \to \overline{k}$ is regular, then for every $y \in V$ there is some neighbourhood W for which

$$g(z_1, ..., z_{n+1}) = \frac{P(z_1, ..., z_{n+1})}{Q(z_1, ..., z_{n+1})} \quad \forall z \in W$$

where $P, Q \in k[Y_1, ..., Y_{n+1}]$ and $Q(z) \neq 0$. Define the rational functions in $k(X_1, ..., X_n)$ as follows

$$R(X_1, ..., X_n) \coloneqq P\left(X_1, ..., X_n, F(X_1, ..., X_n)^{-1}\right)$$
$$S(X_1, ..., X_n) \coloneqq Q\left(X_1, ..., X_n, F(X_1, ..., X_n)^{-1}\right)$$

Let $U \coloneqq \theta^{-1}(W)$. Then evidently $R(x) = P(\theta(x))$ and $S(x) = Q(\theta(x))$ for all $x \in U$. In particular we deduce that $S(\theta^{-1}(y)) \neq 0 \implies S \neq 0$ and that $g \circ \theta$ is regular at $\theta^{-1}(y)$. The converse is straightforward so we deduce that θ is an isomorphism. ∎

When considering regular maps over an open subset of an affine variety, there are two plausible definitions which we show are equivalent.

Proposition 6.1.39 (Quasi-affine Morphisms)
Let $X \subset \mathbb{A}_k^n$, $Y \subset \mathbb{A}_k^m$ be affine varieties, $U \subset X\left(\overline{k}\right)$ an open subset and $\phi : U \to Y\left(\overline{k}\right)$ a function. Then the following are equivalent

a) $\pi_j \circ \phi$ is regular for all $j = 1, ..., m$.
b) ϕ determines a regular morphism $(U, \mathcal{O}_X|_U) \to (Y, \mathcal{O}_Y)$ of spaces with functions.

If $V \subset Y\left(\overline{k}\right)$ is an open subset and $\phi(U) \subset V$ then ϕ determines a regular morphism $(U, \mathcal{O}_X|_U) \to (V, \mathcal{O}_Y|_V)$.

Proof. b) $\implies$ a) follows trivially because $\pi_j : Y\left(\overline{k}\right) \to \overline{k}$ is regular.

a) $\implies$ b) Firstly we observe that for any $f \in \mathcal{O}_X(X)$ for which $D(f) \subset U$ there is a commutative diagram

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$$\begin{array}{ccc} D(f) & \xrightarrow{\phi|_{D(f)}} & Y\left(\overline{k}\right) \\ \sim\downarrow\theta & \nearrow_{\tilde{\phi}_f} & \\ V(\mathfrak{a}^e + (1 - X_{n+1}F)) & & \end{array}$$

where θ is defined in (6.1.38) and $\tilde{\phi}_f(x_1, ..., x_{n+1}) = \phi(x_1, ..., x_n)$. By definition $\tilde{\phi}_f$ is a regular morphism of affine varieties and therefore determines a regular morphism of spaces of functions (6.1.37). As θ is a regular isomorphism we find that $\phi|_{D(f)}$ is a regular morphism of spaces of functions. As the $D(f)$ form a base on the topology of U we may conclude from (4.3.27) that ϕ is a regular morphism of spaces with functions. ■

Proposition 6.1.40
Let X, Y, Z be affine varieties with $U \subset X$ and $V \subset Y$ open subsets. Suppose that $\phi : U \to V$ is regular and $\psi : V \to Z$ is regular, then $\psi \circ \phi : U \to Z$ is regular.

Corollary 6.1.41
Let X be an affine variety, $U \subset X$ an open subset and $f \in \mathcal{O}_X(U)$. Then the set

$$D(f) \coloneqq \{x \in U \mid f(x) \neq 0\}$$

is open and $f|_{D(f)}$ is invertible.

Proof. By (6.1.39) we may regard f as a regular morphism $(U, \mathcal{O}_X|_U) \to \mathbb{A}^1_k\left(\overline{k}\right)$, and it is in particular continuous. Then $\{0\} = V(X_1)$ is a closed subset of $\mathbb{A}^1_k\left(\overline{k}\right)$ and so $D(f) = U \setminus f^{-1}(\{0\})$ is an open subset. It is clear by the definition of the structure sheaf that $f|_{D(f)}$ is invertible. ■

Corollary 6.1.42 (Affine Mapping Property)
Let X, Y be affine varieties. Then there are bijections

$$\mathrm{Mor}(X, Y) \xrightarrow{\sim} \mathrm{Mor}((X, \mathcal{O}_X), (Y, \mathcal{O}_Y)) \xrightarrow{\sim} \mathrm{AlgHom}_k(\mathcal{O}_Y(Y), \mathcal{O}_X(X))$$

where the second map is given by $\phi^\sharp_Y$. Furthermore these are natural in either X or Y, and preserve isomorphisms.

Proof. The first map is well-defined by (6.1.39) and evidently injective. The composite map is bijective by (6.1.30), which shows the first map is surjective and hence bijective. We conclude trivially the second map is also a bijection. ■

Proposition 6.1.43 (Equaliser is Closed)
Let $\phi, \psi : X \to Y$ be regular morphisms of affine varieties. Then the equaliser

$$\left\{x \in X\left(\overline{k}\right) \;\middle|\; \phi(x) = \psi(x)\right\}$$

is closed.

Proof. By applying the projections we may reduce to the case $Y = \mathbb{A}^1_k$. Then $\phi - \psi$ is regular and the equaliser is the inverse image of $\{0\}$ which is evidently closed in $\mathbb{A}^1_k$. As regular maps are continuous then we are done. ■

Proposition 6.1.44
Let $Y = V(\mathfrak{a})$ be an affine variety and $Z \coloneqq V(\mathfrak{b})$ a closed subvariety for $\mathfrak{a} \lhd k[Y]$. Then the space of functions $(Z, \mathcal{O}_Z)$ coincides with the notion of a closed subspace (4.3.21).

Further the inclusion map $i : Z \hookrightarrow Y$ is regular and the induced map

$$i^\sharp_Y : \mathcal{O}_Y(Y) \longrightarrow \mathcal{O}_Z(Z)$$

is a surjective homomorphism with kernel precisely $\rho(\mathfrak{b})$ where $\rho : k[Y] \xrightarrow{\sim} \mathcal{O}_Y(Y)$ is the canonical isomorphism.

Proof. We have a surjective homomorphism $\pi : k[Y] \to k[Z] \coloneqq k[Y]/\mathfrak{b}$ with kernel precisely $\mathfrak{b}$. Suppose $f \in \mathcal{O}_Y(U)$ is a regular map, then we wish to show that $f|_{U \cap Z}$ is regular. But if f is given locally by a ratio $\frac{g}{h}$ then $f|_{U\cap Z} = i^\sharp_U(f)$ is given locally by the ratio $\frac{\pi(g)}{\pi(h)}$. And because π is surjective this means $i^\sharp_Y$ is surjective. Furthermore we have a commutative diagram

$$\begin{array}{ccc} \mathcal{O}_Y(Y) & \xrightarrow{i^\sharp} & \mathcal{O}_Z(Z) \\ \sim\uparrow\rho & & \sim\uparrow\rho \\ k[Y] & \xrightarrow{\pi} & k[Z] \end{array}$$

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from which the properties of $i_Y^\sharp$ follows immediately.

Recall that the Zariski topology on Z coincides with the subspace topology (4.3.21), and we have shown the notion of regular function coincides with the closed subspace definition (4.3.23). Therefore the topology on Z and structure sheaves $\mathcal{O}_Z$ coincide. ∎

Proposition 6.1.45
Let $\phi : X \to Y$ be a regular morphism of affine varieties. Then ϕ is a closed immersion (4.3.23) if and only if $\phi^\sharp$ is surjective.

Proof. Suppose that $\phi^\sharp$ is surjective and let $Z \coloneqq V\big(\ker\big(\phi^\sharp\big)\big)$ be a closed subset of Y. Then $i : Z \hookrightarrow Y$ is regular and we have a commutative diagram

$$\begin{array}{ccc} \mathcal{O}_Y(Y) & \xrightarrow{\phi^\sharp} & \mathcal{O}_X(X) \\ \downarrow{\scriptstyle i^\sharp} & \nearrow_{\tilde{\phi}^\sharp} & \\ \mathcal{O}_Z(Z) & & \end{array}$$

By (6.1.44) $\ker\big(i^\sharp\big) = \ker\big(\phi^\sharp\big)$, so $\tilde{\phi}^\sharp$ exists and is an isomorphism by (3.3.56). By (6.1.42) there exists a corresponding regular morphism $\tilde{\phi} : X \to Z$ which is an isomorphism which satisfies $i \circ \tilde{\phi} = \phi$. By the criteria in (4.3.23) this means ϕ is a closed immersion.

Conversely if ϕ is a closed immersion then by definition $\phi = i \circ \tilde{\phi}$ for $i : Z \hookrightarrow Y$ the inclusion of a closed subset where $\tilde{\phi}$ is a regular isomorphism. By (6.1.42) $\tilde{\phi}^\sharp \circ i^\sharp = \phi^\sharp$ and $\tilde{\phi}^\sharp$ is an isomorphism, which shows that $\phi^\sharp$ is surjective. ∎

6.1.4 Dominant Morphisms

Definition 6.1.46
Let $\phi : X \to Y$ be a regular morphism of affine varieties. Then we say it is

- ***finite*** *if $\phi_Y^\sharp : \mathcal{O}_Y(Y) \to \mathcal{O}_X(X)$ is module-finite (equivalently integral)*
- ***dominant*** *if $\phi\big(X\big(\overline{k}\big)\big)$ is dense in $Y\big(\overline{k}\big)$*

Proposition 6.1.47 (Criteria for dominant morphisms)
A regular morphism $\phi : X \to Y$ is dominant if and only if $\phi_Y^\sharp : \mathcal{O}_Y(Y) \to \mathcal{O}_X(X)$ is injective. In particular when both X and Y are integral then this induces a field extension

$$\phi_\star : k(Y) \hookrightarrow k(X)$$

Proof. Recall $\phi(X)$ is dense precisely when the closure $\overline{\phi(X)} = Y$. Then

$$\begin{aligned} \ker\big(\phi_Y^\sharp\big) = I_k(\phi(X)) &\implies V_{\overline{k}}\big(\ker\big(\phi^\sharp\big)\big) = V_{\overline{k}}(I_k(\phi(X))) = \overline{\phi(X)} \\ &\implies \sqrt{\ker(\phi^\sharp)} = I_k\big(\overline{\phi(X)}\big) \end{aligned}$$

by (6.1.2) and (6.1.13). If $\ker\big(\phi_Y^\sharp\big) = 0$ then clearly $\overline{\phi(X)} = Y$. Conversely if this holds then

$$\sqrt{\ker\big(\phi_Y^\sharp\big)} = I_k\big(Y\big(\overline{k}\big)\big) = I_k(V_{\overline{k}}(0)) = \sqrt{(0)} = (0)$$

as $\mathcal{O}_Y(Y)$ is reduced, whence $\ker\big(\phi_Y^\sharp\big) = 0$. ∎

6.1.5 Dimension

Definition 6.1.48 (Dimension)
Let X be an affine variety. Then we define the dimension to be

$$\dim X \coloneqq \dim X\big(\overline{k}\big) = \dim X_0\big(\overline{k}\big)$$

where this is the Krull Dimension (4.1.81) of the $\overline{k}$-rational points with the k-Zariski topology. This is the supremum of dimension over all irreducible components by (4.1.83).

We say X is of ***pure dimension*** *n if all irreducible components have the same dimension n. Note an integral variety is always of pure dimension.*

We say $X = V(\mathfrak{a})$ is an ***affine curve*** *if it is of pure dimension 1.*

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Proposition 6.1.49 (Dimension of subspace)
Let $X = V(\mathfrak{a})$ be an affine variety and $Y = V(\mathfrak{b})$ an affine subvariety Then

a) $\dim Y = \dim(\mathfrak{b}) = \dim k[X]/\mathfrak{b} = \dim k[Y]$
b) $\operatorname{codim}(Y, X) = \operatorname{ht}(\mathfrak{b})$

Further when $Y = V(\mathfrak{p})$ is integral then $\operatorname{codim}(Y, X) = \dim k[X]_{\mathfrak{p}}$.

Proof. a) follows from (6.1.23).

b) follows similarly by observing that the definition of ideal height (3.21.1) is dual to the topological definition of codimension (4.1.81). The last statement follows from (3.21.4). ■

Corollary 6.1.50
Let $X = V(\mathfrak{a})$ be an affine variety then $\dim X = \dim k[X]$. *In particular* $\dim \mathbb{A}_k^n = n$.

Proof. The first statement is a consequence of (6.1.49) in the case $\mathfrak{b} = (0)$. The dimension for $\mathbb{A}_k^n$ then follows (3.34.28). ■

As we showed when studying the Krull dimension of affine algebras, the lattice of irreducible subsets is particularly well behaved.

Proposition 6.1.51 (Biequidimensional Algebraic Sets)
Let $X = V(\mathfrak{a})$ be an affine variety. Then $X\left(\overline{k}\right)$ is quasi-biequidimensional.

Further X is equidimensional (e.g. integral) iff it is biequidimensional. In this case for every closed subset $Y = V(\mathfrak{b})$ the codimension formula is satisfied

$$\dim X = \dim Y + \operatorname{codim}(Y, X)$$

or in algebraic terms

$$\dim k[X] = \dim \mathfrak{b} + \operatorname{ht}(\mathfrak{b})$$

Proof. We've observed that the lattice of radical (resp. prime) ideals of $k[X]$ is isomorphic to the lattice of closed (resp. irreducible) subsets of $X\left(\overline{k}\right)$. Therefore by (3.34.35) $X\left(\overline{k}\right)$ is quasi-biequidimensional.

In the equidimensional case, the codimension formula follows from (4.1.87) and the algebraic version from (6.1.49).■

In the integral case we recover the "classical" field-theoretic version of dimension

Proposition 6.1.52 (Dimension of Function Field)
Let $X = V(\mathfrak{p})$ be an integral affine variety. Then

$$\dim X = \dim k[X] = \operatorname{trdeg}(\operatorname{Frac}(k[X])/k)$$

Proof. We have already shown that $\dim X = \dim k[X]$. The second equality follows from Noether Normalisation (3.34.28). ■

Proposition 6.1.53
Let X be an affine variety and $Z \subset X$ a closed subset. Then the following are equivalent

a) $\dim Z = 0$
b) Z *is finite*

More precisely

$$Z = \overline{\{x_1\}} \cup ... \cup \overline{\{x_n\}}$$

for a finite number of points $x_1, ..., x_n \in Z$.

Proof. This follows from (4.1.91) and (6.1.18). ■

Proposition 6.1.54 (Hypersurfaces)
Let $X = V(\mathfrak{a})$ be an affine variety and $f \in k[X]$. Define the ***hypersurface*** $Y := V(f)$ *then* $\dim Y = n - 1$ *(and Y is equidimensional).*

Conversely if $k[X]$ is a UFD and $Y \subset X$ is an irreducible closed subset of dimension 1, then Y is a hypersurface.

Proof. **TODO.** ■

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6.1.6 Local Rings

Definition 6.1.55 (Local Ring)
*Let X be an affine variety and $W \subset X\left(\overline{k}\right)$ be an irreducible subset. Then we define the **local ring** at W to be*

$$\mathcal{O}_{X,W} \coloneqq \varinjlim_{U \cap W \neq \emptyset} \mathcal{O}_X(U)$$

It is a local ring with unique maximal ideal

$$\mathfrak{m}_{X,W} = \{(U,\sigma) \mid \sigma(x) = 0 \quad \forall x \in W\}.$$

In the case $W = \{(x)\}$ then we write it as $\left(\mathcal{O}_{X,x}, \mathfrak{m}_{X,x}\right)$.

Proof. First of all the family of open sets such that $U \cap W \neq \emptyset$ is directed by reverse inclusion precisely because W is irreducible, see (4.1.60).

Any section $\sigma \in \mathcal{O}_X(U)$ is continuous with respect to the cofinite topology on k. Therefore $D(\sigma) \coloneqq \sigma^{-1}(k \setminus \{0\}) \subset U$ is an open set and $\sigma \notin \mathfrak{m}_{X,W} \implies D(\sigma) \cap W \neq \emptyset$. Then evidently $[(D(\sigma), \sigma^{-1})]$ is a multiplicative inverse for (U,σ) and by (...) $\mathcal{O}_{X,W}$ is a local ring with unique maximal ideal $\mathfrak{m}_{X,W}$. ∎

Corollary 6.1.56 (Field of Rational Functions)
*Let X be an integral affine variety. Then $\mathcal{O}_{X,X} =: k(X)$ is a field, which we denote as **field of rational functions**.*

Proof. Evidently $\mathfrak{m}_{X,X} = (0)$. Explicitly (V,σ) has inverse $(D(\sigma), \sigma^{-1})$. ∎

Proposition 6.1.57 (Local Ring is Localization of Coordinate Ring)
Let $X = V(\mathfrak{a})$ be an affine variety and $W \subset X\left(\overline{k}\right)$ an irreducible subset with prime ideal $\mathfrak{M}_{X,W} \coloneqq I(W)$. For all such W we have a canonical local isomorphism

$$\begin{array}{ccc} \mathcal{O}_X(X) & \longrightarrow & \mathcal{O}_X(X)_{\mathfrak{M}_{X,W}} \\ & \searrow^{\rho_W} & \downarrow \sim \\ & & \mathcal{O}_{X,W} \end{array}$$

$$\frac{f}{g} \mapsto \left[\left(D(g), \frac{f}{g}\right)\right]$$

The inverse image of $\mathfrak{m}_{X,W}$ is $\mathfrak{M}_{X,W}$ and $\mathfrak{M}_{X,W}\mathcal{O}_X(X)_{\mathfrak{M}_{X,W}}$ respectively. This induces isomorphisms

$$\mathcal{O}_X(X)/\mathfrak{M}_{X,W} \xrightarrow{\sim} k\left(\mathfrak{M}_{X,W}\mathcal{O}_X(X)_{\mathfrak{M}_{X,W}}\right) \xrightarrow{\sim} k\left(\mathfrak{m}_{X,W}\right)$$

Further we have

$$\dim \mathcal{O}_{X,W} = \operatorname{ht}\left(\mathfrak{M}_{X,W}\right) = \operatorname{codim}(W,X) = \dim X - \dim W$$

In particular

$$\dim \mathcal{O}_{X,x} = \dim X$$

Proof. This is a formal consequence of generic facts regarding localization and direct limits

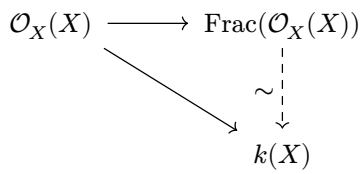
$$\mathcal{O}_X(X)_{\mathfrak{p}} \overset{(3.5.44)}{\cong} \varinjlim_{f \notin \mathfrak{p}} \mathcal{O}_X(X)_f \cong \varinjlim_{D(f) \cap W \neq \emptyset} \mathcal{O}_X(D(f)) \overset{(2.6.127)}{\cong} \varinjlim_{U \cap W \neq \emptyset} \mathcal{O}_X(U)$$

We may demonstrate this more directly. For it is surjective because the principal open sets form a basis and by (6.1.26), observing that $f \notin \mathfrak{p} \iff D(f) \cap V(\mathfrak{p}) \neq \emptyset$. Suppose $\frac{g}{f}$ and $\frac{g'}{f'}$ have the same image then there is some h such that $D(h) \subseteq D(ff') = D(f) \cap D(f')$ and $h \notin \mathfrak{p}$ such that $\frac{g}{f} = \frac{g'}{f'}$ are equal in $k[X]_h$ and a-fortiori in $k[X]_{\mathfrak{p}}$. Therefore the map is injective as required.

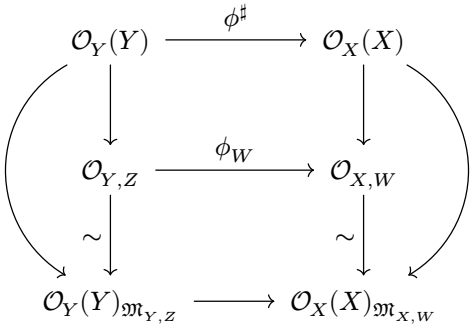
The first two dimension identities follow from (6.1.49) the third from (6.1.51). The final identity follows from (6.1.53).∎

Corollary 6.1.58
Let X be an irreducible variety then there is a canonical isomorphism with the field of rational functions

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Proposition 6.1.59
Let $\phi : X \to Y$ be a regular morphism of affine varieties. Suppose $W \subset X$ is irreducible and $Z \subset \overline{\phi(W)}$ for some irreducible $Z \subset Y$. Then the stalk map $\phi_W : \mathcal{O}_{Y,Z} \to \mathcal{O}_{X,Z}$ (see (4.3.5)) makes the following diagram commute

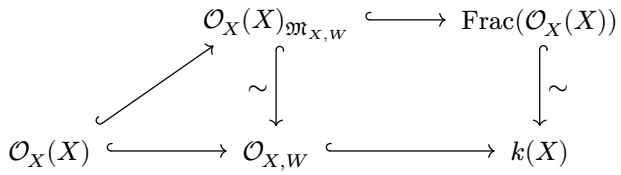


where the bottom arrow is the localisation map induced by $\phi^\sharp$ (3.5.22). In fact it is the unique map making either square commute.

Proposition 6.1.60
Let X be an integral affine variety, $W = V(\mathfrak{M}_{X,W})$ an irreducible subset. Then there are canonical injections

$$\mathcal{O}_X(X) \hookrightarrow \mathcal{O}_{X,W} \hookrightarrow k(X)$$

and the restriction maps are injective. Further there is a commutative diagram



which shows $k(X)$ is a quotient field for $\mathcal{O}_{X,W}$ and $\mathcal{O}_X(X)$.

Proof. We first show the restriction maps are injective. Suppose $\sigma|_V = 0$. Note that V is dense (4.1.59), σ is continuous (6.1.34) with respect to the k-Zariski topology on $\overline{k}$ so that $\sigma^{-1}(\{0\})$ is closed. By definition this contains V and therefore also $\mathrm{cl}_U(V) = \mathrm{cl}_X(V) \cap U = U$. We conclude $\sigma = 0$ and the restriction maps are injective. The same argument shows that the remaining maps are injective.

The vertical maps are isomorphisms by (6.1.57), and we may verify directly the diagram commutes. ∎

Therefore if it's convenient to do so we may identify $\mathcal{O}_X(X)$ and $\mathcal{O}_{X,W}$ as subrings of $k(X)$.

Proposition 6.1.61
Let X be an affine variety and $W \subset X$ an irreducible closed subset. Then the minimal primes of $\mathcal{O}_{X,W}$ are in bijection with the irreducible components of X containing W.

$\mathcal{O}_{X,W}$ is an integral domain if and only if W lies in a unique irreducible component.

Proof. By (6.1.57) and (3.5.42) the minimal primes of $\mathcal{O}_{X,W}$ correspond to minimal primes of $k[X]$ contained in $\mathfrak{M}_{X,W}$. These correspond to irreducible components of X containing $\overline{W}$ (and hence W) by (6.1.19).

As $\mathcal{O}_{X,W}$ is reduced, it is an integral domain iff it has a unique minimal prime ideal (3.3.67). ∎

Proposition 6.1.62
Let X be an affine variety, R an algebraic k-algebra (3.34.25) and $\phi : \mathcal{O}_{X,x} \to R$ a k-algebra homomorphism. Then $\phi^{-1}(\mathfrak{m}_R) = \mathfrak{m}_{X,x}$ and therefore factors through $k(\mathfrak{m}_{X,x})$

This in particular this includes the case R is an algebraic field extension.

Proof. See (3.34.27). ∎

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6.1.7 Rational Points

Proposition 6.1.63 (K-rational points)
Let $X = V(\mathfrak{a})$ be an affine variety and K an algebraic field extension. Then there are natural bijections

$$\begin{aligned} X(K) &\longleftrightarrow \mathrm{AlgHom}_k(\mathcal{O}_X(X), K) \\ &\longleftrightarrow \{(\mathfrak{M}, \phi) \mid \mathfrak{M} \in \mathrm{Specm}(\mathcal{O}_X(X)) \quad \phi : \mathcal{O}_X(X)/\mathfrak{M} \hookrightarrow K\} \\ &\longleftrightarrow \left\{(x, \phi) \;\middle|\; x \in X_0\left(\overline{k}\right) \quad \phi : k\left(\mathfrak{m}_{X,x}\right) \hookrightarrow K\right\} \\ &\longleftrightarrow \left\{(x, \phi) \;\middle|\; x \in X_0\left(\overline{k}\right) \quad \phi : \mathcal{O}_{X,x} \to K\right\} \end{aligned}$$

Proof. The first correspondence is simply (6.1.7) and (6.1.26).

For the second correspondence consider a k-algebra homomorphism $\phi : \mathcal{O}_X(X) \to K$. By (3.34.27) we have $\ker(\phi) =: \mathfrak{M}$ is maximal, which induces a homomorphism $\mathcal{O}_X(X)/\mathfrak{M} \hookrightarrow K$. Evidently this correspondence is injective and surjective.

By (6.1.11) there is a bijection $\mathrm{Specm}(\mathcal{O}_X(X)) \longleftrightarrow X_0\left(\overline{k}\right)$. Further there is an isomorphism $\mathcal{O}_X(X)/\mathfrak{M}_{X,x} \cong k\left(\mathfrak{m}_{X,x}\right)$ (6.1.57) from which the third correspondence is well-defined and bijective.

The final correspondence follows from (6.1.62) because all such homomorphisms factor through $k\left(\mathfrak{m}_{X,x}\right)$. ∎

6.1.8 Generic Points

When considering irreducible subsets $W \subset X$ there is another way of looking at these in terms of "generic points".

Definition 6.1.64 (Generic Point)
Let $X = V(\mathfrak{a})$ be an affine variety and $(\xi) \in X(\Omega)$ for some extension field Ω/k. Then we may define the irreducible subset of $X\left(\overline{k}\right)$

$$W_\xi := V_{\overline{k}}(\mathfrak{p}_\xi) = \left\{(x) \in X\left(\overline{k}\right) \;\middle|\; \forall f \in k[X](f(\xi) = 0 \implies f(x) = 0)\right\}$$

There are canonical isomorphisms

$$k\left(\mathfrak{m}_{X,W}\right) \xrightarrow{\sim} k[X]_{\mathfrak{p}}/\mathfrak{p}k[X]_{\mathfrak{p}} \xrightarrow{\sim} \mathrm{Frac}(k[X]/\mathfrak{p}) \xrightarrow{\sim} k(\xi)$$

*We say that (ξ) is a **generic point** corresponding to the irreducible closed subset W_ξ. Moreover every irreducible subset is of this form for we may simply consider $\Omega := \mathrm{Frac}(k[X]/\mathfrak{p})$ and $\xi = \left(\overline{X}_1, ..., \overline{X}_n\right)$.*

*If $(x) \in W_\xi$ then we say that (x) is a **specialization** of (ξ) and this induces the specialization homomorphism*

$$\begin{array}{ccc} k[X]/\mathfrak{p}_\xi & \longrightarrow & k[X]/\mathfrak{m}_x \\ \wr & & \wr \\ k[\xi] & \dashrightarrow & k[x] \end{array}$$

If $W_\xi = X$ then we say simply (ξ) is a generic point.

6.1.9 Tangent Space and Non-Singular Points

We propose two definitions for the tangent space and show that under mild technical conditions they are naturally isomorphic. The latter definition may be identified geometrically.

Definition 6.1.65
*Let X be an affine variety and let $W \subset X$ be an irreducible subset. We define the **cotangent space** at W to be the $k\left(\mathfrak{m}_{X,W}\right)$-vector space*

$$T^\star_W X := \mathfrak{m}_{X,W}/\mathfrak{m}^2_{X,W}$$

*and the **tangent space** to be the $k\left(\mathfrak{m}_{X,W}\right)$-vector space*

$$T_W X := \mathrm{Der}_k\left(\mathcal{O}_{X,W}, k\left(\mathfrak{m}_{X,W}\right)\right)$$

For points this has a concrete interpretation

Proposition 6.1.66 (Concrete interpretation of Tangent Space)
Let $X = V(\mathfrak{a}) \subset \mathbb{A}^n_k$ be an affine variety and $x \in X\left(\overline{k}\right)$. Then there are natural $k(x)$-module isomorphisms

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$$\begin{array}{lll} T_xX & \tilde{\to} \operatorname{Der}_k(k[X], k(x)) & \tilde{\to} \left\{ v \in k(x)^n \;\middle|\; \sum_{i=1}^n v_i \frac{\partial F}{\partial X_i}(x) = 0 \quad \forall F \in \mathfrak{a} \right\} \\ D & \to & \left(D\left(\overline{X}_1\right), ..., D\left(\overline{X}_n\right) \right) \end{array}$$

where we have used the identification $k\left(\mathfrak{m}_{X,x}\right) \cong k(x)$ *and the* k*-algebra homomorphism* $k[X] \to k\left(\mathfrak{m}_{X,x}\right)$*. Suppose* $\mathfrak{a} = \langle F_1, ..., F_m \rangle$ *then the right hand side is the kernel of the following* $k(x)$*-module homomorphism*

$$\begin{pmatrix} \frac{\partial F_1}{\partial X_1}(x) & \cdots & \frac{\partial F_1}{\partial X_n}(x) \\ \vdots & \ddots & \vdots \\ \frac{\partial F_m}{\partial X_1}(x) & \cdots & \frac{\partial F_m}{\partial X_n}(x) \end{pmatrix} : k(x)^n \to k(x)^m$$

In particular

$$\dim_{k\left(\mathfrak{m}_{X,x}\right)} T_xX = n - \operatorname{rk}\left(\frac{\partial F_i}{\partial X_j}(x) \right)$$

Proof. Recall by (6.1.57) and (6.1.64) that $\mathcal{O}_{X,x} \cong k[X]_{\mathfrak{M}_{X,x}}$ and $k\left(\mathfrak{m}_{X,x}\right) \cong k(x)$ so we have isomorphisms

$$\operatorname{Der}_k\left(\mathcal{O}_{X,x}, k\left(\mathfrak{m}_{X,x}\right)\right) \cong \operatorname{Der}_k\left(k[X]_{\mathfrak{M}_{X,x}}, k(x)\right) \overset{(3.25.10)}{\cong} \operatorname{Der}_k(k[X], k(x))$$

and the remaining isomorphism is from (3.34.41) ∎

Proposition 6.1.67 (Dimension Formula of Tangent Space)
Let $X = V(\mathfrak{a}) \subset \mathbb{A}_k^n$ *be an* integral *affine variety,* $W \subset X$ *an irreducible subset. Suppose that* $\mathfrak{a} = \langle F_1, ..., F_m \rangle$ *and let* $\xi := \left(\overline{X}_1, ..., \overline{X}_n\right) \in k\left(\mathfrak{m}_{X,W}\right)^n$ *be the image of the coordinate functions. Then we have the dimension formula*

$$\dim T_WX = \dim \operatorname{Der}\left(k[X], k\left(\mathfrak{m}_{X,W}\right)\right) = n - \operatorname{rk}\left(\frac{\partial F_i}{\partial X_j}(\xi) \right)$$

Suppose that $Z \subset W$ *is an irreducible subset then we have the inequality*

$$\dim T_ZX \geq \dim T_WX$$

and

$$\dim T_WX \geq \dim T_XX = \operatorname{Der}(k(X))$$

which equals $\dim X$ *when* $k(X)$ *is separably generated (e.g.* k *is perfect).*

Proof. We have isomorphisms

$$\begin{aligned} \dim T_WX := & \quad \operatorname{Der}_k\left(\mathcal{O}_{X,W}, k\left(\mathfrak{m}_{X,W}\right)\right) \\ \overset{(6.1.57)}{\to} & \quad \operatorname{Der}_k\left(\mathcal{O}_X(X)_{\mathfrak{M}_{X,W}}, k\left(\mathfrak{m}_{X,W}\right)\right) \\ \overset{(3.25.10)}{\to} & \quad \operatorname{Der}_k\left(\mathcal{O}_X(X), k\left(\mathfrak{m}_{X,W}\right)\right) \\ \overset{(6.1.26)}{\to} & \quad \operatorname{Der}_k\left(k[X], k\left(\mathfrak{m}_{X,W}\right)\right) \end{aligned}$$

and the rank formula follows from (3.34.41) and (3.3.135).

Let ζ be the corresponding generic point of Z, then because $\mathfrak{M}_{X,W} \subset \mathfrak{M}_{X,Z}$ we may construct a k-algebra homomorphism $k[\xi] \to k[\zeta]$. So by the determinant criteria of rank (3.8.43) we deduce that

$$\operatorname{rk}\left(\frac{\partial F_i}{\partial X_j}(\zeta) \right) \leq \operatorname{rk}\left(\frac{\partial F_i}{\partial X_j}(\xi) \right)$$

from which the inequality follows.

Observe that $k\left(\mathfrak{m}_{X,X}\right) =: k(X)$ and $T_XX \tilde{\to} \operatorname{Der}_k(k[X], k(X)) \tilde{\to} \operatorname{Der}_k(k(X))$. When $k(X)$ is separably generated this equals $\dim X$ by (3.34.44). ∎

Proposition 6.1.68 (Duality between Tangent and Cotangent Space)
Let X *be an affine variety and* $W \subset X$ *be an irreducible subset. There is a canonical* $k\left(\mathfrak{m}_{X,W}\right)$*-module homomorphism*

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$$T_W X \to (T^\star_W X)^\vee$$

Under the condition that $k(\mathfrak{m}_{X,W})/k$ *is separably generated (e.g.* k *is perfect) then this map is an isomorphism. In particular* $\dim T_W X = \dim T^\star_W X$ *by (3.3.109).*

Proof. This follows directly from (3.34.50) by considering the local ring $A \coloneqq \mathcal{O}_{X,W}$ with maximal ideal $\mathfrak{m}_{X,W}$. ∎

Lemma 6.1.69
Let A *be a reduced ring with minimal prime ideal* $\mathfrak{p}$ *and prime ideal* $\mathfrak{q} \supset \mathfrak{p}$ *containing no other minimal prime ideals. Suppose that*

$$\phi : A \to B$$

is a surjective ring homomorphism with $\ker(\phi) = \mathfrak{p}$*. Then* ϕ *induces an isomorphism of rings*

$$\tilde{\phi} : A_\mathfrak{q} \to B_{\phi(\mathfrak{q})}$$

which is the unique morphism commuting with ϕ*.*

Proof. The map $\tilde{\phi}$ exists and has kernel $\mathfrak{p}A_\mathfrak{q}$ by (3.5.5). We need simply to show that $\mathfrak{p}A_\mathfrak{q} = (0)$. By (3.5.42) it is the unique minimal prime ideal of $A_\mathfrak{q}$. As $A_\mathfrak{q}$ is reduced then we are done by (3.3.46). ∎

Proposition 6.1.70
Let X *be an affine variety,* W *an irreducible subset contained in precisely one irreducible component* X_α*. Then there is a canonical isomorphism*

$$\begin{aligned} \mathcal{O}_{X,W} &\xrightarrow{\sim} \mathcal{O}_{X_\alpha,W} \\ (U,\sigma) &\to \left(U \cap X_\alpha, \sigma|_{X_\alpha \cap U}\right) \end{aligned}$$

and in particular an isomorphism of tangent spaces

$$T_W X \xrightarrow{\sim} T_W X_\alpha$$

Proof. The regular map $i : X_\alpha \hookrightarrow X$ corresponds to a ring homomorphism $i^\sharp$ with kernel $\mathfrak{M}_{X,X_\alpha}$ by (6.1.44). By (...) we have a commutative diagram

$$\begin{array}{ccc} \mathcal{O}_X(X) & \xrightarrow{i^\sharp} & \mathcal{O}_{X_\alpha}(X_\alpha) \\ \downarrow & & \downarrow \\ \mathcal{O}_{X,W} & \xrightarrow{i_W} & \mathcal{O}_{X_\alpha,W} \\ \downarrow \sim & & \downarrow \sim \\ \mathcal{O}_X(X)_{\mathfrak{M}_{X,W}} & \longrightarrow & \mathcal{O}_{X_\alpha}(X_\alpha)_{\mathfrak{M}_{X_\alpha,W}} \end{array}$$

and by (6.1.69) we deduce that the stalk map is an isomorphism. ∎

Definition 6.1.71 (Non-Singular Points)
Let X *be an affine variety and* $W \subset X$ *be an irreducible subset. We say that* X *is* ***non-singular*** *at* W *if*

a) W *is contained in only one irreducible component* X_α*, and*
b) $\dim T_W X = \dim \mathrm{Der}_k(k(X_\alpha))$

Note that by (6.1.61) a) $\iff \mathcal{O}_{X,W}$ *is an integral domain.*

Recall by (6.1.67) that $\dim T_W X = \dim T_W X_\alpha \geq \dim \mathrm{Der}_k(k(X_\alpha))$ *and in the case of* k *perfect this equals* $\dim X_\alpha$*.*

We may now show the following important result

Proposition 6.1.72 (Non-Singular Point)
Let $X = V(\mathfrak{a}) \subset \mathbb{A}^n_k$ *be an affine variety then the non-singular points form an open dense subset of* X*.*

Proof. We may reduce to the case of X irreducible. Let $\Omega \coloneqq k(X)$ and $\xi \coloneqq \left(\overline{X}_1, ..., \overline{X}_n\right) \in X(\Omega)$ and suppose $\mathfrak{p} = \langle F_1, ..., F_m \rangle \lhd k[X_1, ..., X_n]$ is the ideal defining X. Then

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$$\mathrm{Der}(\Omega) \overset{(3.25.10)}{=} \mathrm{Der}(k[X], \Omega) \overset{(3.34.41)}{\cong} \ker\left(\frac{\partial F_i}{\partial X_j}(\xi)\right)$$

For $x \in X\left(\overline{k}\right)$ there is a k-algebra homomorphism $k[\xi] \to k[x]$ and so by the determinant criteria of rank (3.8.43) we see

$$\mathrm{rk}\left(\frac{\partial F_i}{\partial X_j}(x)\right) \leq \mathrm{rk}\left(\frac{\partial F_i}{\partial X_j}(\xi)\right)$$

Suppose the rank of the right hand side is r and consider the r-minors $g_1, ..., g_m \in k[X]$. Then by the same result at least one is non-zero and the set of non-singular points is given by

$$\bigcup_p D(g_p)$$

which is non-empty by (6.1.27). ∎

Proposition 6.1.73 (Non-Singular Variety)
Let X be an integral affine variety. Then the following are equivalent

a) X is non-singular at all points $x \in X\left(\overline{k}\right)$
b) X is non-singular at all irreducible subsets $W \subset X\left(\overline{k}\right)$

*In this case we say that X is a **non-singular affine variety**.*

Proof. Evidently $b) \implies a)$ because $T_x X = T_{\overline{\{x\}}} X$. Conversely given any $x \in W$ we have by (6.1.67)

$$T_x X \geq T_W X \geq \dim \mathrm{Der}(k(X))$$

whence they are both equal. ∎

Definition 6.1.74 (Regular Point)
*Let X be an affine variety and $W \subset X$ an irreducible subset. Then we say that W is **regular** if*

a) W is contained in precisely one irreducible component X_α, and
b) $\mathcal{O}_{X,W}$ is a regular local ring (i.e. $\dim T^\star_W X = \dim \mathcal{O}_{X,W}$)

*We say X is **regular** if it is regular at all irreducible subsets.*

Fortunately these concepts are typically equivalent.

Proposition 6.1.75 (Regular = Non-Singular in Perfect Case)
Let $X = V(\mathfrak{a}) \subset \mathbb{A}^n_k$ an affine variety with k perfect and $W \subset X$ an irreducible subset. Then

$$\dim T^\star_W X = \dim T_W X$$

and the following are equivalent

a) X is regular at W
b) X is non-singular at W
c) $\dim T_W X = \dim \mathcal{O}_{X,W} = \mathrm{codim}(W, X)$

When $\mathrm{codim}(W, X) = 1$ this is equivalent to $\mathcal{O}_{X,W}$ being a discrete valuation ring.

Proof. Recall by (6.1.57) that $\dim \mathcal{O}_{X,x} = \dim X$ and by definition $T^\star_x X = \mathfrak{m}_{X,x}/\mathfrak{m}^2_{X,x}$. Further by definition (x) is regular iff $\dim T^\star_x X = \dim \mathcal{O}_{X,x}$. Finally by (6.1.68) $\dim T^\star_x X = \dim T_x X$ so we see $a) \iff c)$.

Similarly $b) \iff c)$ is essentially by definition. The final statement follows from (3.30.2). ∎

Example 6.1.76 (Simple Points of a Plane Curve)
The canonical example is a plane curve $k[X] = k[X,Y]/(F(X,Y))$ which has dimension 1 (...). Given $(x) \in X(K)$ we see that

$$T_x X = \left\{ (\alpha, \beta) \in k(x)^2 \;\middle|\; 0 = \alpha \frac{\partial F}{\partial X}(x) + \beta \frac{\partial F}{\partial Y}(x) \right\}$$

*Clearly this has dimension 1 (in which case (x) is a **simple point**) unless both the partial derivatives vanish at (x) in which case it has dimension 2.*

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Clearly $F(X,Y) = Y^2 - X^3$ has a non-simple point at $(0,0)$ but $F(X,Y) = Y - X^2$ has simple points everywhere.

6.1.10 Zeta Function over Finite Fields

For this section let $k = \mathbb{F}_p$ with algebraic closure $\overline{k}$. Then for every $n \geq 1$ by (...) $\overline{k}$ has a unique subfield k_n of degree n, and order p^n. Furthermore we have the following properties

a) $k_m \subseteq k_n \iff [k_m : k] \mid [k_n : k] \iff m \mid n$
b) $[k_n : k_m] = \frac{n}{m}$
c) k_n/k_m is Galois with the group of automorphisms generated by ϕ^m for all integers $m \mid n$, and the Galois group has order $\frac{n}{m}$.
d) every finite subextension is of this form

Lemma 6.1.77
Let $x_1, ..., x_N \in \overline{k}$. Then

$$k(x_1, ..., x_N) = k_n$$

where

$$n \coloneqq \operatorname{lcm}_i [k(x_i) : k]$$

Proof. By definition $[k(x_i) : k] \mid n$ whence $k(x_i) \subseteq k_n$ by the previous observations, and therefore $k(x_1, ..., x_N) \subseteq k_n$.

Similarly as $k(x_i) \subset k(x_1, ..., x_N)$ we have $[k(x_i) : k] \mid [k(x_1, ..., x_N) : k]$. Then by definition $n \mid [k(x_1, ..., x_N) : k]$ which shows $k_n \subset k(x_1, ..., x_N)$. ∎

The following result shows that the number of rational points over a finite field may be characterized by algebraic properties of the coordinate ring $k[X]$. This is essentially because maximal ideals correspond to Galois orbits of rational points and we may count rational points by summing over Galois orbits.

Proposition 6.1.78 (Counting Rational Points)
Let $X = V(\mathfrak{a}) \subset \mathbb{A}_k^N$ be an affine variety. Then for all integers $n \geq 1$ we have the following relation

$$\#X(k_n) = \sum_{d \mid n} d \times \#\{\mathfrak{m} \mid \dim_k k(\mathfrak{m}) = d\}$$

Proof. For $(x) \in X\left(\overline{k}\right)$, then by the previous discussion

$$(x) \in X(k_n) \iff k(x) \subseteq k_n \iff \dim_k k(x) \mid n$$

This shows that

$$\begin{aligned} \#X(k_n) &= \sum_{d \mid n} \#\{(x) \in X(k_n) \mid \dim_k k(x) = d\} \\ &= \sum_{d \mid n} \#\{(x) \in X(k_d) \mid \dim_k k(x) = d\} \end{aligned} \tag{*}$$

By (6.1.11) we have a bijection

$$X(k_d)/\operatorname{Gal}(k_d/k) \xrightarrow{\sim} \{\mathfrak{m} \mid \dim_k k(\mathfrak{m}) \mid d\}$$

As we have an isomorphism $k(x) \cong k(\mathfrak{m}_x)$ this restricts to a bijection

$$\{(x) \in X(k_d) \mid \dim_k k(x) = d\}/\operatorname{Gal}(k_d/k) \xrightarrow{\sim} \{\mathfrak{m} \mid \dim_k k(\mathfrak{m}) = d\}$$

We claim that the action of $\operatorname{Gal}(k_d/k) = \langle \phi \rangle$ is free. For suppose $(x) \in X(k_d)$ such that $\dim_k k(x) = d$ and $\phi^r(x_i) = x_i$ for $i = 1, ..., N$ and $0 < r \leq d$. Then by (3.18.108) we have $x_i \in k_r$ whence $\dim_k k(x) \leq r$. This shows we must have $r = d$ and in particular only the identity automorphism fixes (x).

Therefore we see that

$$\#\{(x) \in X(k_d) \mid \dim_k k(x) = d\} = \#\operatorname{Gal}(k_d/k) \times \#\{\mathfrak{m} \mid \dim_k k(\mathfrak{m}) = d\}$$

From (3.18.111) we recall $\operatorname{Gal}(k_d/k)$ is cyclic of order d so we may combine this with (*) to obtain the required result. ∎

Proposition 6.1.79 (Zeta function of an affine variety over a finite field)
Formally as elements of the power series ring $\mathbb{Q}[[T]]$ we have

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$$Z(X,T) := \prod_{\mathfrak{m}\in \operatorname{Specm}(k[X])} \left(1 - T^{\deg(\mathfrak{m})}\right)^{-1} = \exp\left(\sum_{m=1}^{\infty} \#\frac{X(k_m)}{m} T^m\right)$$

Proof. Define

$$b_d := \#\{\mathfrak{m} \mid \dim_k k(\mathfrak{m}) = d\}$$

Let $Z(X,T)$ be the right hand side then

$$\begin{aligned}
\log(Z(X,T)) &= \sum_{m=1}^{\infty} \#X(k_m)\frac{T^m}{m} \\
&= \sum_{m=1}^{\infty} \sum_{d|m} (d \times b_d)\frac{T^m}{m} \\
&= \sum_{d=1}^{\infty} b_d \sum_{r=1}^{\infty} \frac{T^{rd}}{r} \\
&= -\sum_{d=1}^{\infty} b_d \log\left(1 - T^d\right)
\end{aligned}$$

■

Example 6.1.80
For $X(k) = k^n$ we have $\#X(k_m) = p^{mn}$. Then

$$Z(X,T) = \exp\left(\sum_{m=1}^{\infty} \frac{p^{mn}T^m}{m}\right) = \exp(-\log(1 - p^n T)) = \frac{1}{1 - p^n T}$$

6.1.11 Base Change

Definition 6.1.81 (Affine Base Change)
Let $X = V(\mathfrak{a}) \subset \mathbb{A}^n_k$ be an affine variety over k and K/k a field extension. Consider the canonical ring homomorphism

$$i_{kK} : k[X_1, ..., X_n] \hookrightarrow K[X_1, ..., X_n]$$

and let $\mathfrak{a}^e := \mathfrak{a}K[X_1, ..., X_n]$ denote the extension under this map.

*Further define the **base change** of X with respect to K/k to be the affine variety*

$$X_K := V\left(\sqrt{\mathfrak{a}^e}\right) \subset \mathbb{A}^n_K$$

Proposition 6.1.82 (Coordinate Ring of Base Change)
Let $X = V(\mathfrak{a}) \subset \mathbb{A}^n_k$ be an affine variety over k and K/k a field extension. There is a canonical K-algebra isomorphism

$$\begin{aligned}
\left(k[X] \otimes_k K\right)_{\text{red}} &\xrightarrow{\sim} K[X_K] \\
\overline{F} \otimes \lambda &\to \lambda \cdot \overline{i_{kK}(F)}
\end{aligned}$$

Proof. By (3.4.42) we have K-algebra isomorphisms

$$k[X] \otimes_k K \xrightarrow{\sim} (k[X_1, ..., X_n] \otimes_k K)/\mathfrak{a}^e \xrightarrow{\sim} K[X_1, ..., X_n]/\mathfrak{a}^e$$

By (3.34.4) and (3.34.2) this determines a unique K-algebra isomorphism

$$\left(k[X] \otimes_k K\right)_{\text{red}} \xrightarrow{\sim} \left(K[X_1, ..., X_n]/\mathfrak{a}^e\right)_{\text{red}} \xrightarrow{\sim} K[X_1, ..., X_n]/\sqrt{\mathfrak{a}^e} =: K[X_K]$$

■

It may be interesting to understand the cases when $\mathfrak{a}^e$ remains radical under the base change.

Proposition 6.1.83
Let $X = V(\mathfrak{a})$ be an affine variety. Then the extension $\mathfrak{a}^e$ under $A \to A \otimes_k K$ remains radical under any of the following conditions

a) $k[X]$ is geometrically reduced
b) K/k is geometrically reduced (e.g. separably generated)

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c) k is perfect.

Proof. By (3.4.42) we have an isomorphism

$$K[X_1, ..., X_n]/\mathfrak{a}^e \xrightarrow{\sim} k[X] \otimes_k K$$

so that $\mathfrak{a}^e$ is radical iff $k[X] \otimes_k K$ is reduced. Then we may reduce to results on geometrically reduced algebras. ∎

We are particularly interested when the variety remains integral (equivalently irreducible) under base change.

Definition 6.1.84 (Geometrically Integral)
*Let $X = V(\mathfrak{a})$ be an affine variety over k. Then we say X is **geometrically integral** if X_K is integral for all field extensions K. By (...) this is equivalent to $k[X]$ being geometrically irreducible.*

Proposition 6.1.85 (Criteria to be Geometrically Integral)
Let $X = V(\mathfrak{a}) \subset \mathbb{A}^n_k$ be an integral affine variety. Then the following are equivalent

a) X is geometrically integral
b) $X_{\overline{k}}$ is integral
c) $X_{k'}$ is integral for every finite separable extension k'/k
d) $k[X]$ is geometrically irreducible.
e) $k[X] \otimes \overline{k}$ is irreducible
f) $k[X] \otimes k'$ is irreducible for every finite separable extension k'/k.
g) $k(X)/k$ is relatively separably closed

Proof. We have *a)* $\iff$ *d)*, *b)* $\iff$ *e)* and *c)* $\iff$ *f)* are straightforward since a ring A is irreducible iff A_{red} is integral.

For *d)* $\iff$ *e)* $\iff$ *f)* $\iff$ *g)* we may use (3.35.41). ∎

Proposition 6.1.86
Let $X = \mathbb{A}^n_k$. Then there is a canonical isomorphism $X_K \cong \mathbb{A}^n_K$. In particular X is geometrically integral.

Proposition 6.1.87
Let $X \subset \mathbb{A}^n_k$ be an affine variety, K/k a field extension and R a K-algebra. Then there is a bijection

$$\begin{array}{ccc} X\big(R_{(k)}\big) & \xrightarrow{\quad\sim\quad} & X_K(R) \\ \downarrow \sim & & \downarrow \sim \\ \mathrm{AlgHom}_k\big(k[X], R_{(k)}\big) & \xrightarrow{\sim} & \mathrm{AlgHom}_K(K[X_K], R) \\ \phi & \to & \big(\lambda \cdot \overline{F} \to \lambda \cdot \phi\big(\overline{i_{kK}(F)}\big)\big) \end{array}$$

for $F \in k[X_1, ..., X_n]$, where the vertical arrows are given by (...).

Corollary 6.1.88
Let X be an affine variety and $L/K/k$ a tower of extensions, then we may identify $X(L)$ with $X_K(L)$.

Proposition 6.1.89
Let $X = V(\mathfrak{a})$ an affine variety and $\overline{k}/K/k$ a tower of extensions. Then after identifying $X\big(\overline{k}\big)$ with $X_K\big(\overline{k}\big)$ we have the following properties

a) U open in $X\big(\overline{k}\big) \implies U$ open in $X_K\big(\overline{k}\big)$,
b) There is a well-defined K-algebra homomorphism

$$(\mathcal{O}_X(U) \otimes_k K)_{\text{red}} \to \mathcal{O}_{X_K}(U)$$

$$f \otimes \lambda \to \lambda \cdot f$$

which is an isomorphism when $U = X$,
c) sets of the form $D\big(\sum_{i=1}^n \lambda_i f_i\big)$ are open in $X_K\big(\overline{k}\big)$ and form a base for the Zariski topology.

Proof.

a) A closed subset of $X\big(\overline{k}\big)$ is of the form $Z = V(\mathfrak{b})$ for $\mathfrak{b} \subset \mathfrak{a}$. Then evidently $Z = V(\mathfrak{b}^e)$ and so it is closed as a subset of $X_K\big(\overline{k}\big)$
b) By definition we have an open cover $U = \bigcup_{i=1}^n U_i$ and polynomials $g_i, h_i \in k[X]$ such that $f|_{U_i} = \frac{g_i}{h_i}$. This shows that $\lambda \cdot f$ is also regular. The map is an isomorphism on global sections by (6.1.82).

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c) By (6.1.39) $f \coloneqq \sum_{i=1}^{n} \lambda_i f_i$ is continuous. As $\{0\}$ is a closed subset of $\overline{k}$ in the Zariski (co-finite) topology this shows that $D(f)$ is open. These form a base because $D(fg) = D(f) \cap D(g)$, and specifically for the Zariski topology by (6.1.15) and b).

■

6.1.12 Valuation Rings on the Function Field

Proposition 6.1.90
Let K/k be a field extension and R a valuation ring of K/k. Then there exists a valuation ring $R' \subset R$ such that $k(\mathfrak{m}_{R'})/k$ is algebraic and $\mathfrak{m}_{R'} = \mathfrak{m}_R \cap R'$.

Proof. By (3.24.11) the identity map $1 : k \to \overline{k}$ extends to a ring homomorphism $S \to \overline{k}$ where $(S, \mathfrak{m}_S)$ is a valuation ring of $k(\mathfrak{m}_R)/k$. Considering the quotient map $\pi : R \to k(\mathfrak{m}_R)$ let $R' \coloneqq \pi^{-1}(S)$. The restriction map

$$\pi' : R' \to S \to k(\mathfrak{m}_S)$$

is surjective, and so the kernel $\mathfrak{m}_{R'} \coloneqq \ker(\pi') = \mathfrak{m}_R \cap R'$ is a maximal ideal of R' and $k(\mathfrak{m}_{R'}) \cong k(\mathfrak{m}_S)$. Observe that $R' \subseteq R$ and $\mathfrak{m}_R \subseteq \mathfrak{m}_{R'}$. If $x \notin R$ then $x^{-1} \in \mathfrak{m}_R \implies x^{-1} \in R$ by (3.24.2). If $x \in R \setminus R'$ then $\pi(x) \notin S \implies \pi(x^{-1}) \in S \implies x^{-1} \in R'$. Therefore we conclude that R' is a valuation ring with maximal ideal $\mathfrak{m}_{R'}$ and algebraic residue field as required. ■

The following result was proven in the case of a finitely generated k-algebra (3.34.28), where equality holds. In general only the inequality holds by considering for example the case $\dim K = 0$. The proof below is taken from [Rey11].

Lemma 6.1.91
Let R be a subring of a finitely generated field extension K/k. Then $\dim R \leq \operatorname{trdeg}(K/k)$.

Proof. Consider a chain of prime ideals

$$\mathfrak{p}_0 \subsetneq ... \subsetneq \mathfrak{p}_n \subset R$$

Choose $x_i \in \mathfrak{p}_i \setminus \mathfrak{p}_{i-1}$ and define $A \coloneqq k[x_1, ..., x_n] \subset R$ with $K' \coloneqq \operatorname{Frac}(A) \subset K$. Then there is a chain of prime ideals of A

$$\mathfrak{p}_0 \cap A \subsetneq ... \subsetneq \mathfrak{p}_n \cap A$$

whence $n \leq \dim A \overset{(3.34.28)}{=} \operatorname{trdeg}(K'/k) \overset{(3.18.140)}{\leq} \operatorname{trdeg}(K/k)$. Taking supremum over n we reach the required result. ■

Lemma 6.1.92
Let $R \subseteq S$ be proper valuation rings of K with $\dim R < \infty$. Then $\dim S \leq \dim R$, with equality iff $R = S$.

Proof. Firstly we claim that $\mathfrak{m}_S \subseteq \mathfrak{m}_R$. For $x \in \mathfrak{m}_S \implies x^{-1} \notin S \implies x^{-1} \notin R \implies x \in \mathfrak{m}_R$.

Secondly if $R \subsetneq S$ then $\mathfrak{m}_S \subsetneq \mathfrak{m}_R$. For given $x \in S \setminus R$ we have $x^{-1} \in \mathfrak{m}_R \implies x^{-1} \in S \implies x^{-1} \in S^\star \implies x^{-1} \notin \mathfrak{m}_S$.

Let $\mathfrak{p}_0 \subsetneq ... \subsetneq \mathfrak{p}_n$ be a chain of prime ideals in S. As these are contained in $\mathfrak{m}_S$, they are also contained in $\mathfrak{m}_R$ and therefore R. Therefore this constitutes a chain of prime ideals in R from which we deduce the inequality. Further if $R \subsetneq S$ then we may always extend the chain by at least one prime ideal, namely $\mathfrak{m}_R$. ■

Proposition 6.1.93
Let $X = V(\mathfrak{a}) \subset \mathbb{A}^n_k$ be an integral affine variety, and $\mathcal{O}_X(X) \subset R \subset k(X)$ a valuation ring. Then there exists a point $x \in X\big(\overline{k}\big)$ such that R dominates $\mathcal{O}_{X,x}$. Further $\dim R \leq \dim X$ with equality iff $\mathcal{O}_{X,x} = R$ as subrings of $k(X)$.

Proof. By (6.1.90) we may assume that $k(\mathfrak{m}_R)/k$ is algebraic. Choose any embedding $k(\mathfrak{m}_R)/k \hookrightarrow \overline{k}/k$ (3.18.61). Then there is a composite k-algebra homomorphism

$$\mathcal{O}_X(X) \hookrightarrow R \to k(\mathfrak{m}_R) \hookrightarrow \overline{k}$$

Let x_i be the image of the affine coordinates $\overline{X}_i$ under this homomorphism. The image $k[x_1, ..., x_n]$ is a (finite) field extension of k by (3.18.44). Then under this map the image of f is $f(x)$, so the kernel is precisely $\mathfrak{M}_{X,x} = \mathfrak{m}_R \cap \mathcal{O}_X(X)$. Suppose that $f \notin \mathfrak{M}_{X,x}$ then $f \notin \mathfrak{m}_R \implies f^{-1} \in R$. In particular we conclude that $\mathcal{O}_{X,x} = \mathcal{O}_X(X)_{\mathfrak{M}_{X,x}} \subset R$. Further $\mathfrak{m}_{X,x} = \mathfrak{M}_{X,x}\mathcal{O}_{X,x} \subseteq \mathfrak{m}_R R \subseteq \mathfrak{m}_R$, so R dominates $\mathcal{O}_{X,x}$ by (3.24.4).

The final statement follows from (6.1.92) and $\dim X = \dim \mathcal{O}_{X,x}$ (6.1.57). ■

6.1.13 Products of Affine Varieties

In what follows we let A_{red} denote the reduced ring (resp k-algebra) $A/N(A)$.

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Proposition 6.1.94
There is a canonical isomorphism

$$\begin{aligned} \psi : k[X_1,...,X_n] \otimes_k k[Y_1,...,Y_m] &\xrightarrow{\sim} k[X_1,...,X_n,Y_1,...,Y_m] \\ F(X_1,...,X_n) \otimes G(Y_1,...,Y_m) &\to \quad F(X_1,...,X_n)G(Y_1,...,Y_m) \end{aligned}$$

Let $\mathfrak{a} \lhd k[X_1,...,X_n]$ *and* $\mathfrak{b} \lhd k[Y_1,...,Y_m]$ *be ideals. Then there is a commutative diagram*

$$\begin{array}{ccc} k[X_1,...,X_n] \otimes_k k[Y_1,...,Y_m] & \xrightarrow[\sim]{\psi} & k[X_1,...,X_n,Y_1,...,Y_m] \\ \downarrow{\scriptstyle \pi_1 \otimes \pi_2} & & \downarrow \\ \frac{k[X_1,...,X_n]}{\mathfrak{a}} \otimes_k \frac{k[Y_1,...,Y_m]}{\mathfrak{b}} & \xrightarrow[\sim]{\overline{\psi}} & \frac{k[X_1,...,X_n,Y_1,...,Y_m]}{\mathfrak{a}^e+\mathfrak{b}^e} \end{array}$$

$$\overline{F} \otimes \overline{G} \to \overline{F(X_1,...,X_n)G(Y_1,...,Y_m)}$$

where the bottom arrow is uniquely determined and an isomorphism. The nilradical of the right hand side is $\sqrt{\mathfrak{a}^e+\mathfrak{b}^e}/(\mathfrak{a}^e+\mathfrak{b}^e)$ *and so there is a corresponding isomorphism*

$$\left(\frac{k[X_1,...,X_n]}{\mathfrak{a}} \otimes_k \frac{k[Y_1,...,Y_m]}{\mathfrak{b}}\right)_{\text{red}} \xrightarrow{\sim} \frac{k[X_1,...,X_n,Y_1,...,Y_m]}{\sqrt{\mathfrak{a}^e+\mathfrak{b}^e}}$$

Proof. By (3.4.37) there is a unique k-algebra homomorphism ψ such that $\psi(F \otimes 1) = F$ and $\psi(1 \otimes G) = G$, whence $\psi(F \otimes G) = FG$. Similarly there is an inverse morphism ϕ such that $\phi(X_i) = X_i \otimes 1$ and $\phi(Y_j) = 1 \otimes Y_j$. Evidently $\phi(F(X_1,...,X_n)) = F \otimes 1$ and $\phi(G(Y_1,...,Y_m)) = 1 \otimes G$. This shows that $\phi \circ \psi = \mathbf{1}$. Similarly $(\psi \circ \phi)(X_i) = X_i$ and $(\psi \circ \phi)(Y_j) = Y_j$. This shows that $\psi \circ \phi = \mathbf{1}$ as required.

There is a commutative diagram

$$\begin{array}{ccc} k[X_1,...,X_n] & \xrightarrow{i} & k[X_1,...,X_n] \otimes_k k[Y_1,...,Y_m] \\ & \searrow & \wr \downarrow \psi \\ & & k[X_1,...,X_n,Y_1,...,Y_m] \end{array}$$

By (3.4.42) $\mathfrak{a} \otimes k[Y_1,...,Y_m]$ is the extension of $\mathfrak{a}$ under i. As the diagram commutes we may show that $\psi(\mathfrak{a} \otimes k[Y_1,...,Y_m]) = \mathfrak{a}^e$ and similarly $\psi(k[X_1,...,X_n] \otimes \mathfrak{b}) = \mathfrak{b}^e$. Therefore by (3.3.51)

$$\psi(\mathfrak{a} \otimes k[Y_1,...,Y_m] + k[X_1,...,X_n] \otimes \mathfrak{b}) = \mathfrak{a}^e + \mathfrak{b}^e$$

By (3.4.29) $\ker(\pi_1 \otimes \pi_2) = \mathfrak{a} \otimes k[Y_1,...,Y_m] + k[X_1,...,X_n] \otimes \mathfrak{b}$ therefore $\overline{\psi}$ exists and is an isomorphism by (...). ∎

Lemma 6.1.95
Let $X = V(\mathfrak{a}) \subset \mathbb{A}_k^n$ *and* $Y = V(\mathfrak{b}) \subset \mathbb{A}_k^m$*. Consider the affine variety* $\widetilde{X \times Y} := V\left(\sqrt{\mathfrak{a}^e+\mathfrak{b}^e}\right) \subset \mathbb{A}_k^{n+m}$*. The projection maps*

$$\begin{aligned} \pi_X &: \widetilde{X \times Y}\left(\overline{k}\right) \to X\left(\overline{k}\right) \\ \pi_Y &: \widetilde{X \times Y}\left(\overline{k}\right) \to Y\left(\overline{k}\right) \end{aligned}$$

are regular and there is an isomorphism

$$\begin{aligned} \left(\mathcal{O}_X(X) \otimes_k \mathcal{O}_Y(Y)\right)_{\text{red}} &\xrightarrow{\sim} \mathcal{O}_{\widetilde{X \times Y}}\left(\widetilde{X \times Y}\right) \\ f \otimes g \to \quad & (f \circ \pi_X) \cdot (g \circ \pi_Y) \end{aligned}$$

where $\pi_X : \widetilde{X \times Y}\left(\overline{k}\right) \to X\left(\overline{k}\right)$ *and* $\pi_Y : \widetilde{X \times Y}\left(\overline{k}\right) \to Y\left(\overline{k}\right)$ *are the obvious regular projection maps.*

Proof. It's straightforward to show that π_X and π_Y are regular using the definition (6.1.29). By (6.1.94) and (6.1.26) there is a unique morphism making the following diagram commute which must be an isomorphism, and we simply need to check it is of the required form.

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$$\begin{array}{ccc} (k[X] \otimes_k k[Y])_{\text{red}} & \xrightarrow{\sim} & k\big[\widetilde{X \times Y}\big] \\ \downarrow \sim & & \downarrow \sim \\ (\mathcal{O}_X(X) \otimes_k \mathcal{O}_Y(Y))_{\text{red}} & \dashrightarrow & \mathcal{O}_{\widetilde{X \times Y}}\big(\widetilde{X \times Y}\big) \end{array}$$

Given $f \in k[X]$ and $g \in k[Y]$ where $f = \overline{F}$ and $g = \overline{G}$. Then the image of $f \otimes g$ is simply $h := \overline{F(X_1, ..., X_n)G(Y_1, ..., Y_n)}$ and

$$h(x_1, ..., x_n, y_1, ..., y_m) = f(x_1, ..., x_n)g(y_1, ..., y_m)$$

from which the result follows. ∎

Proposition 6.1.96 (Affine Product)
Let $X = V(\mathfrak{a}) \subset \mathbb{A}^n_k$ and $Y = V(\mathfrak{b}) \subset \mathbb{A}^m_k$ be affine varieties. Then there is a bijection

$$\begin{aligned} X\big(\overline{k}\big) \times Y\big(\overline{k}\big) &\xrightarrow{\sim} \widetilde{X \times Y}\big(\overline{k}\big) \\ (x, y) \to \ & (x_1, ..., x_n, y_1, ..., y_m) \end{aligned}$$

Define $\big(X\big(\overline{k}\big) \times Y\big(\overline{k}\big), \mathcal{O}_{X \times Y}\big)$ as a local space of functions such that this bijection is a regular isomorphism.

The projection maps $\pi_X : X \times Y \to X$ and $\pi_Y : X \times Y \to Y$ are regular and for every affine variety Z we have a natural bijection

$$\begin{aligned} \text{Mor}((Z, \mathcal{O}_Z), (X \times Y, \mathcal{O}_{X \times Y})) &\xrightarrow{\sim} \text{Mor}((Z, \mathcal{O}_Z), (X, \mathcal{O}_X)) \times \text{Mor}((Z, \mathcal{O}_X), (Y, \mathcal{O}_Y)) \\ f \to \ & ((\pi_X \circ f), (\pi_Y \circ f)) \end{aligned}$$

Proof. Evidently it is enough to prove all the same properties for $\widetilde{X \times Y}$, as then they follow immediately for $X \times Y$ by composing with θ, which is by construction a regular isomorphism. We have natural isomorphisms

$$\begin{aligned} \text{Mor}\big(Z, \widetilde{X \times Y}\big) &\xrightarrow{\sim} \text{AlgHom}_k\big(\mathcal{O}_{\widetilde{X \times Y}}\big(\widetilde{X \times Y}\big), \mathcal{O}_Z(Z)\big) \\ &\xrightarrow{\sim} \text{AlgHom}_k\big((\mathcal{O}_X(X) \otimes_k \mathcal{O}_Y(Y))_{\text{red}}, \mathcal{O}_Z(Z)\big) \\ &\xrightarrow{\sim} \text{AlgHom}_k(\mathcal{O}_X(X) \otimes_k \mathcal{O}_Y(Y), \mathcal{O}_Z(Z)) \\ &\xrightarrow{\sim} \text{AlgHom}_k(\mathcal{O}_X(X), \mathcal{O}_Z(Z)) \times \text{AlgHom}_k(\mathcal{O}_Y(Y), \mathcal{O}_Z(Z)) \\ &\xrightarrow{\sim} \text{Mor}(Z, X) \times \text{Mor}(Z, Y) \end{aligned}$$

by (6.1.95) and (6.1.30). We may verify that the image of $1_{\widetilde{X \times Y}}$ is (π_X, π_Y) and so the result follows from naturality in Z. ∎

Proposition 6.1.97 (Affine Varieties are Separated)
Let $f : X \to Y$ be a regular map of affine varieties. Then the graph

$$\Gamma_f(K) := \{(x, f(x)) \mid x \in X(K)\}$$

is a closed subset of $(X \times Y)(K)$. In particular the diagonal

$$\Delta_X := \{(x, x) \mid x \in X(K)\}$$

is a closed subset of $(X \times X)(K)$.

Proof. Let $\overline{X}_1, ..., \overline{X}_n$ and $\overline{Y}_1, ..., \overline{Y}_m$ be the coordinate functions of $k[X \times Y]$. Further we may suppose that $f = \big(\overline{F}_1, ..., \overline{F}_m\big)$ for $F_j \in k[X_1, ..., X_n]$. Then Γ_f is the zero locus of the ideal

$$\big\langle \overline{Y}_1 - F_1\big(\overline{X}_1, ..., \overline{X}_n\big), ..., \overline{Y}_m - F_m\big(\overline{X}_1, ..., \overline{X}_n\big) \big\rangle$$

∎

Proposition 6.1.98 (Criteria for Product to be Integral)
Let X, Y be integral affine varieties such that either X or Y is geometrically *integral (e.g. if k is algebraically closed). Then $X \times Y$ is integral.*

Proof. See (3.35.42). ∎

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6.1.14 Affine Curves

Definition 6.1.99
*An **affine curve** is an integral affine variety of dimension* 1.

For example an irreducible polynomial $F(X,Y)$ yields an affine curve and is known as a plane curve. The singular points are precisely the common roots of

$$F, \frac{\partial F}{\partial X}, \frac{\partial F}{\partial Y}$$

in $\overline{k}^2$. If there are no such roots then the affine curve is non-singular.

Proposition 6.1.100 (Characterisation Regular Point on a curve)
Let $X = V(\mathfrak{a})$ be an affine curve and $x \in X\left(\overline{k}\right)$. Then the following are equivalent

a) x is regular
b) $\mathcal{O}_{X,x}$ is a discrete valuation ring
c) $\mathcal{O}_{X,x}$ is an integrally closed domain

When k is perfect then this is equivalent to x being a non-singular point.

Proof. Note that by assumption X is integral, so $\mathcal{O}_{X,x}$ is an integral domain by (6.1.61).

a)-c) then follows directly from (3.30.2) and the fact $\dim \mathcal{O}_{X,x} = \dim X = 1$ (6.1.57). The final statement follows from (6.1.75). ∎

Proposition 6.1.101
Let $X = V(\mathfrak{a})$ be a regular affine curve. Then there exists a bijection

$$X_0 \coloneqq X\left(\overline{k}\right) / \operatorname{Aut}\left(\overline{k}/k\right) \longleftrightarrow \{\mathcal{O}_X(X) \subset R \subsetneq k(X) \mid R \text{ valuation ring}\}$$

Proof. The map $x \to \mathcal{O}_{X,x}$ is well-defined by (6.1.100), (6.1.9) and (6.1.60). If $\mathcal{O}_{X,x} = \mathcal{O}_{X,y}$ then $\mathcal{O}^\star_{X,x} = \mathcal{O}^\star_{X,y} \implies \mathfrak{m}_{X,x} = \mathfrak{m}_{X,y} \implies \mathfrak{M}_{X,x} = \mathfrak{M}_{X,y} \implies x \sim y$ by (6.1.9) so the map is injective.

To show surjectivity, observe by (6.1.93) there exists $x \in X\left(\overline{k}\right)$ such that $\mathcal{O}_{X,x} \subseteq R$. Furthermore by repeated application of (6.1.92) we see

$$0 < \dim R \leq \dim \mathcal{O}_{X,x} \overset{(6.1.57)}{=} \dim X = 1$$

This forces $\dim R = \dim \mathcal{O}_{X,x} = 1$ and so $R = \mathcal{O}_{X,x}$ by the same result. ∎

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